

Belief Distortions, Asset Prices, and Unemployment Fluctuations*

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Abstract

This paper studies asset prices and unemployment when firms’ expectations deviate from rationality. Using machine learning forecasts as an objective benchmark, I quantify distortions in survey forecasts of corporate cash flows. Survey forecasts overreact to cash flow news, while machine forecasts do not. These belief distortions explain over 60% of the variation in hiring at both the aggregate and firm levels. Following idiosyncratic cash flow shocks, firms with distorted beliefs overhire relative to objective firms, while their stock returns overshoot and then reverse. A general-equilibrium search model in which firms hold diagnostic beliefs about cash-flow growth reproduces the overreaction and the volatility of asset valuations and unemployment. Beliefs that raise asset valuations also raise the value firms place on new hires, so stock prices and hiring move together.

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1 Introduction

Why are asset prices and unemployment so volatile? The standard search-and-matching (Diamond-Mortensen-Pissarides, DMP) model generates unemployment volatility an order of magnitude smaller than in the data, a disconnect known as the “unemployment volatility puzzle” (Shimer, 2005). Recent rational models explain it with time-varying discount rates inferred from asset prices: recessions bring high risk premia that depress the present value of the long-term cash flows from hiring (Hall, 2017; Borovičková and Borovička, 2017; Kehoe et al., 2023). At the

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same time, growing evidence finds that stock market investors and firm managers are neither fully rational nor efficient at processing information. Yet little work has quantified how much distortions in subjective beliefs can explain the puzzle.

I measure firms’ subjective expectations using consensus survey forecasts from equity analysts and CFOs, which proxy for the beliefs of the managers who make hiring decisions.¹ I then construct a real-time benchmark of efficient expectations using the machine learning methodology of Bianchi et al. (2025) (BLLM, hereafter). I call the deviation between the two, whether from nonrational expectations or informational inefficiency, a *belief distortion*, and quantify how much these distortions explain labor market fluctuations. Whereas BLLM apply this approach to earnings-growth and stock-return expectations, I extend it to show that the same distortions account for large fluctuations in unemployment and hiring.

I document four results showing that belief distortions in expected cash flows, the deviation between survey and machine forecasts of corporate earnings, account for much of the variation in aggregate unemployment, firm-level hiring, and predictable stock returns. First, survey forecasts systematically overreact to cash flow news while machine forecasts do not, confirming the presence of distortions. Second, this distortionary component explains a large share of variation in both labor markets and asset prices: in the aggregate time series, 90% of vacancy filling rates and 68% of unemployment; in the cross-section, 66% of the dispersion in hiring rates, 50% of profits per worker, and 62% of stock returns. Third, following positive idiosyncratic shocks, firms with distorted beliefs hire excessively relative to firms with less distorted beliefs, lowering subsequent profits per worker, while their stock returns initially overshoot and then predictably reverse. Finally, a calibrated general-equilibrium search-and-matching model with these distortions reproduces about 88% of observed U.S. unemployment volatility and the cross-sectional dispersion in hiring, against near-zero unemployment volatility under an otherwise equivalent rational-expectations economy.

The paper makes two contributions. First, I go beyond documenting violations of full-information rational expectations (FIRE) to quantifying their importance for labor market outcomes and asset valuations: prior work asks whether survey forecast errors deviate from FIRE, but knowing that a deviation exists does not measure its importance for hiring. Second, belief overreaction offers an alternative to time-varying-discount-rate models, with testable firm-level predictions linking employment decisions to return predictability. Those models price no id-

¹Analyst and managerial earnings forecasts are highly correlated when both are available (Gennaioli et al., 2016), and managers routinely incorporate analyst projections into corporate decisions (Kothari et al., 2016). Analysts also communicate directly with managers and share similar information sets (Brown et al., 2015), making their forecasts a useful measure of the belief environment relevant for hiring decisions.

idiosyncratic risk, which is inconsistent with the evidence here that idiosyncratic shocks move both hiring and stock returns when beliefs are distorted.

The evidence for survey overreaction comes from comparing survey forecasts with machine learning predictions. The algorithm trains a neural network on the same historical information available to survey respondents and deviates from their forecasts only when they made predictable mistakes.² The gap between survey and machine forecasts therefore isolates inefficiencies in belief formation rather than differences in information or hindsight: because the machine conditions on the survey forecasts themselves, it cannot underperform merely for lacking information the survey embeds, and the residual gap reflects predictable mistakes the survey could have avoided in real time. Overreaction shows up in a specific pattern: positive forecast revisions predict negative forecast errors, so upward adjustments are too large and later reversed. Machine forecasts show no such pattern. The wedge between the two, the belief distortion, thus reflects survey overreaction to news, not errors in the machine forecast, and it holds in both the aggregate time series and the firm cross-section, indicating systematic biases rather than measurement error.

Using the firm’s hiring condition from a search-and-matching model, I link the belief distortion to hiring by decomposing the value of hiring into expected cash flows (expected earnings) and discount rates (expected stock returns). This decomposition is analogous to the Campbell and Shiller (1988) present-value identity, which attributes movements in valuation ratios to revisions in expected cash flows versus discount rates. Comparing it under survey and machine forecasts quantifies how much hiring variation arises from biased beliefs.

The decomposition shows that hiring is far more sensitive to cash flow news under subjective than objective beliefs. Under rational expectations with time-varying discount rates, objective discount rates unsurprisingly dominate, explaining 69% of aggregate variation and 60% of cross-sectional dispersion in hiring. Under subjective beliefs, cash flow expectations instead account for 97% and 79%. Crucially, it is not the subjective belief *per se* but its identified distortion, the gap between survey and machine forecasts, that accounts for most of this sensitivity, for both aggregate fluctuations and cross-sectional dispersion. Because the job-finding rate, of which the vacancy filling rate is the matching-function counterpart, accounts for about three-quarters of unemployment fluctuations (Shimer, 2005, 2012), this implies belief distortions explain roughly $90\% \times 3/4 = 68\%$ of aggregate unemployment fluctuations.³ Random noise or analyst bias would

²The machine learning forecast approximates Bayesian updating in a high-dimensional learning problem, with regularization acting as a prior that excludes economically implausible extremes (Martin and Nagel, 2022). This prior reduces the variance of the forecasts, which limits their overreaction and improves their out-of-sample accuracy.

³This is a back-of-the-envelope mapping. Shimer (2012) attributes roughly three-quarters of the variance of

weaken, not strengthen, this link, which operates through firms' subjective asset valuations.

Firm-level responses to idiosyncratic cash flow shocks (measured by earnings growth) differ sharply under objective versus subjective expectations. For firms with near-objective beliefs, hiring should respond only weakly to these transitory, unpriced shocks. Rational models do not preclude hiring responses to idiosyncratic shocks in general; rather, because these particular shocks are highly transitory and hiring is costly to adjust, a near-objective firm anticipates little benefit from expanding before the shock dissipates. Consistent with this, when earnings expectations track the machine benchmark, profits per worker remain stable after idiosyncratic shocks, and stock returns barely respond, as in rational asset pricing, where diversifiable risk is unpriced.

By contrast, firms with distorted beliefs overreact to both aggregate and idiosyncratic shocks in their hiring and stock returns. Firms whose analyst forecasts are overly optimistic relative to the benchmark overhire: inflated beliefs about the value of new workers lead them to expand aggressively, raising labor costs faster than revenues so that profits and profits per worker decline. Their stock prices overshoot as optimism is embedded in valuations, then reverse as earnings disappoint, generating predictable negative abnormal returns. Quantitatively, belief distortions account for 82% of the explanatory power of subjective forecast revisions for contemporaneous profits per worker, versus a much smaller share for objective revisions. Belief distortions thus provide a new mechanism linking firms' hiring responses to unpriced idiosyncratic shocks with predictable patterns in their stock returns.

Finally, I interpret the evidence through a general-equilibrium search-and-matching model with Epstein-Zin preferences and limited asset-market participation, calibrated to the documented overreaction, which lets me compare the observed economy against a counterfactual in which firms hold full-information rational expectations. Firms hold diagnostic beliefs about cash-flow growth, overreacting to recent news while anchoring the long run to the truth; priced by the household's stochastic discount factor, these beliefs inflate the perceived value of hiring in transitory booms and depress it in slumps. Because the wage is rigid, arising from an acyclical belief about the expected re-employment wage, operating leverage turns these movements into large swings in vacancies and unemployment. These distorted cash flow expectations drive excessive fluctuations in the value of hiring, generating the large subjective cash flow component in the variance decompositions.

unemployment fluctuations to the job-finding rate and the remainder to the separation rate. Because the job-finding rate f_t and the vacancy filling rate q_t are both monotone functions of labor market tightness θ_t (with $f_t = q_t\theta_t$), variation in the vacancy filling rate maps into variation in the job-finding margin; the factor 3/4 rescales its variance share into an unemployment share, abstracting from the separation margin.

Related Literature This paper contributes to the literatures on unemployment fluctuations, asset prices, and expectation formation by quantifying how much distorted beliefs relevant to asset prices explain labor market fluctuations.

First, it relates to the literature on the unemployment volatility puzzle in search-and-matching models under the DMP framework (Shimer, 2005).⁴ The model generates enough unemployment volatility only when firms’ responses to shocks are amplified, such as through time-varying discount rates (Hall, 2017; Borovičková and Borovička, 2017; Kehoe et al., 2023).⁵ A related strand attributes unemployment volatility to wage rigidity from credible bargaining (Hall, 2005; Hall and Milgrom, 2008), staggered negotiations (Gertler and Trigari, 2009), or incentive contracts (Gaur et al., 2023). This work assumes firms rationally process information about cash flows and discount rates. I instead show that firms’ overreaction to cash flow news is a distinct mechanism for unemployment fluctuations, producing both the aggregate volatility and the cross-sectional dispersion in hiring that aggregate discount-rate models miss.

A growing literature embeds nonrational expectations in models of labor market frictions (Venkateswaran, 2014; Acharya and Wee, 2020; Mueller et al., 2021; Balleer et al., 2021; Faberman et al., 2022; Menzio, 2023; Pappa et al., 2023; Bhandari et al., 2024; Jäger et al., 2024; Du et al., 2025; Bigio et al., 2025; Ding, 2025; Mitra, 2025; Sheng, 2025).⁶ Bhandari et al. (2024) show that household and firm pessimism can explain unemployment volatility; Bigio et al. (2025) link asset-price and labor-market volatility through heterogeneous beliefs and operational leverage; and Du et al. (2025) document sticky job-finding and separation expectations that lead workers to under-insure in recessions. I add two complementary pieces of evidence: I measure expectational distortions directly against a real-time machine learning benchmark, and I document firm-level overreaction to idiosyncratic shocks in both hiring and stock returns, showing that such shocks shape hiring through belief distortions.

The empirical analysis builds on survey-based evidence on subjective expectations. On the firm side, a large literature documents systematic forecasting biases.⁷ Managers extrapolate re-

⁴See also Hagedorn and Manovskii (2008), Hall and Milgrom (2008), Pissarides (2009), Elsby and Michaels (2013), Kudlyak (2014), Chodorow-Reich and Karabarbounis (2016), and Ljungqvist and Sargent (2017).

⁵Related work that takes an asset pricing perspective on the labor market under rational expectations includes Merz and Yashiv (2007), Donangelo (2014), Belo et al. (2014), Favilukis and Lin (2015), Kuehn et al. (2017), Kilic and Wachter (2018), Mitra and Xu (2019), Donangelo et al. (2019), Kehoe et al. (2019), Liu and Smirnyagin (2021), Belo et al. (2023), and Meeuwis et al. (2023).

⁶Related work links belief distortions to credit cycles (Bordalo et al., 2021; Gulen et al., 2024) and shows that loose credit generates boom-bust dynamics in employment (Blank and Maghzian, 2023). My paper complements this work by showing that distorted cash flow expectations affect hiring directly, over and above their effects through credit conditions.

⁷On the worker side, early work shows that workers’ beliefs including job loss expectations forecast their labor market outcomes (Stephens, 2004).

cent performance, generating predictable forecast errors that influence their decisions (Ben-David et al., 2013; Alti and Tetlock, 2014; Gennaioli et al., 2016; Barrero, 2022; Davis et al., 2025). Greenwood and Hanson (2014) document, and L’Huillier et al. (2025) formalize, how competition neglect leads to overinvestment in booms. Using incentivized forecasts, Bloom et al. (2025) show that firms exhibit overoptimism, predictable errors, and overprecision that distort their responsiveness to fundamentals. Related work links these managerial biases to distorted investment (Ma et al., 2020), internal hurdle rates (Gormsen and Huber, 2025), and sluggish, dispersed inflation expectations (Coibion et al., 2018; Candia et al., 2020). I complement this work by showing that distorted earnings expectations translate into hiring decisions and unemployment fluctuations.

The variance decomposition and learning model in this paper builds on recent work using survey-based expectations to reassess the drivers of asset prices.⁸ The approach adapts the present-value identity framework from Campbell and Shiller (1988) and Cochrane (2008), which attributes variation in valuation ratios to expected cash flows and discount rates. Recent survey-based applications challenge the traditional view that discount-rate variation dominates asset price fluctuations. Bordalo et al. (2024a) find that overreaction in long-term earnings expectations drives return predictability. De La O and Myers (2021) show that subjective expectations of cash flow growth explain most of the variation in valuation ratios, but remain agnostic about whether these expectations systematically over- or under-react to information. I contribute an explicit measure of objective beliefs that quantifies the magnitude and direction of these distortions, and use it to show that belief distortions drive not only asset prices but also real decisions such as hiring, a channel not examined in prior work.

I adopt the machine learning approach of BLLM to measure the magnitude of distortions in subjective beliefs. As in Bianchi et al. (2022) and BLLM, I structure the algorithm so that the machine’s forecasts can differ from the survey forecasts only if the machine finds evidence of predictable mistakes in the survey responses immediately prior to the machine making a true out-of-sample forecast. The method uses tools from machine learning by training Long Short-Term Memory (LSTM) neural networks with recursive re-estimation and hyperparameter tuning (Gu et al., 2020, Cong et al., 2020, Nagel, 2021, van Binsbergen et al., 2022, Bastianello, 2022, Bybee et al., 2024). The resulting forecasts are fully ex-ante, providing empirical counterparts to

⁸Related work includes Timmermann (1993), Barberis et al. (1998), Chen et al. (2013), Greenwood and Shleifer (2014), Collin-Dufresne et al. (2016), Adam et al. (2016), De La O and Myers (2021), Giglio et al. (2021), Nagel and Xu (2022b), Jin and Sui (2022), van Binsbergen et al. (2022), Bordalo et al. (2024a), De La O et al. (2024), Adam and Nagel (2023), Afrouzi et al. (2023), Décaire and Graham (2024), Bastianello et al. (2024), and Chaudhry (2025).

rational expectations. Whereas BLLM use identified belief distortions to explain stock market outcomes, I extend their methodology to the labor market, showing that distorted beliefs have real consequences beyond financial markets.

The rest of the paper proceeds as follows. Section 2 describes the data used in the empirical analysis. Section 3 documents overreaction in survey expectations and compares the predictive performance of machine and survey forecasts. Section 4 presents a search and matching model with belief distortions and derives a decomposition of the vacancy filling rate. Section 5 presents the estimated variance decomposition of the aggregate vacancy filling rate. Section 6 presents cross-sectional evidence motivated by a firm-level extension of the baseline model. Section 7 develops a general-equilibrium search-and-matching model in which measured belief distortions jointly discipline asset prices and unemployment, and shows that it reproduces the decompositions estimated from the data. Section 8 discusses model extensions and robustness checks. Finally, section 9 concludes. Appendix B contains additional results referenced in the main text. A separate Online Appendix (Sections OA through OD) provides supplementary results and technical details supporting the main analysis.⁹

2 Data

This section describes the data used to estimate the time-series and cross-sectional variance decompositions. For each outcome, I measure subjective expectations with survey forecasts $\mathbb{F}_t[\cdot]$ and objective expectations with machine learning forecasts $\mathbb{E}_t[\cdot]$. The estimation sample is quarterly, 2005Q1–2023Q4, the period over which real-time machine forecasts are available out of sample. Throughout, t indexes quarters and i indexes firms.¹⁰

Labor Market Outcomes *Aggregate level.* The vacancy filling rate q_t measures the probability that a posted vacancy is filled in period t :

$$q_t \equiv \frac{\text{Total Hires}}{\text{Total Job Vacancies}} = \frac{f_t U_t}{V_t},$$

where V_t is total job vacancies from Job Openings and Labor Turnover Survey (JOLTS) job openings (available since 2000:12; earlier periods use the help-wanted index, Barnichon, 2010), and U_t is the unemployment level (BLS). The job-finding rate f_t measures the probability that

⁹The Online Appendix is available at <https://doqlee.github.io/files/onlineappendix.pdf>

¹⁰Realized labor-market and earnings series extend back to 1983, and the survey series begin at different dates, with analyst earnings forecasts in 1983, firm-level return forecasts (price targets) in 1999, and the CFO return survey in 2001Q4, so descriptive figures use the longest span available for each. Appendix OC provides additional details, and Figure A.1 reports stylized facts.

an unemployed worker finds a job within the month. Following Shimer (2005), I infer how many people must have exited unemployment each period from the decline in total unemployment after removing very short-term unemployed workers:

$$f_t = 1 - \frac{U_t - U_t^s}{U_{t-1}},$$

where U_t^s is short-term unemployment less than 5 weeks (UEMPLT5). The job separation rate δ_t is from JOLTS. Variables are constructed monthly, aggregated to quarterly averages, and detrended with an HP filter ($\lambda = 10^5$) following Shimer (2005).

Firm level. Since firm-level vacancies are not readily observable, I proxy hiring at the firm level by the hiring rate, defined as net employment growth after accounting for job separations

$$hl_{i,t} \equiv \log\left(\frac{L_{i,t+1} - (1 - \delta_{i,t})L_{i,t}}{L_{i,t}}\right) = \log\left(\frac{\text{Total Hires}}{\text{Total Employment}}\right),$$

where $L_{i,t}$ is employment for firm i , recorded at fiscal year-end and carried forward to quarterly frequency. $\delta_{i,t}$ is the separation rate of firm i 's NAICS2 industry (from JOLTS). Including job separations ensures the hiring rate captures total hires needed to both replace departing workers and expand employment. The firm-level sample includes publicly listed firms with common stocks (share codes 10, 11) on NYSE/AMEX/NASDAQ with Institutional Brokers' Estimate System (IBES) analyst coverage of expected earnings and stock price targets.

Earnings (Realized Cash Flows) *Firm level.* Firm-level earnings $E_{i,t}^*$ are constructed from IBES street earnings per share (EPS), converted to total earnings by multiplying by each firm's shares outstanding (1983Q4–2023Q4). Street earnings exclude one-off items that are not informative for ongoing operations and better proxy expected cash flows (Hillenbrand and McCarthy, 2024). To handle instances where firm earnings may be zero or negative, which would make log transformations undefined, I follow the approach of Vuolteenaho (2002) and define a transformed measure of earnings that is strictly positive. Specifically, each firm is treated as a portfolio consisting of a fraction $(1 - \lambda)$ of equity and a small fraction λ of risk-free assets, such that

$$E_{i,t} = (1 - \lambda)E_{i,t}^* + \lambda r_t^f P_{i,t-1}, \quad \lambda = 0.10,$$

where $P_{i,t-1}$ is lagged equity value and r_t^f is the one-year Treasury bill rate. The risk-free component ensures $E_{i,t} > 0$ for all observations while preserving the cross-sectional variation in earnings. This transformation allows log earnings $e_{i,t} \equiv \log E_{i,t}$ to be well defined even in periods when reported earnings $E_{i,t}^*$ are negative.

Aggregate level. I aggregate firm-level earnings to the S&P 500 level to obtain a time-series for aggregate earnings:

$$E_t = \Omega_t \sum_{i \in x_t} \frac{E_{i,t}^*}{\text{Divisor}_t},$$

where x_t is the set of S&P 500 firms with IBES data, Ω_t is a scaling factor that adjusts for occasionally incomplete IBES coverage of the S&P 500, and Divisor_t is the S&P 500 divisor.

Cash Flow Expectations (Surveys) *Firm level.* Subjective earnings expectations $\mathbb{F}_t[\Delta e_{i,t+h}]$ are obtained from the IBES database, which reports the median consensus forecasts of equity research analysts (1983Q4–2023Q4).

I use analyst forecasts as a proxy for managerial expectations. This approach is supported by three considerations. First, analyst and CFO forecasts of earnings growth are highly correlated (approximately 0.60 at the one-year horizon), indicating broadly consistent beliefs across groups (Gennaioli et al., 2016).¹¹ Second, prior research shows that analyst forecasts are widely followed by managers and investors and influence corporate decisions (Kothari et al., 2016). Third, analysts have access to similar information sets as managers and often communicate directly with them (Brown et al., 2015). Although not produced by managers themselves, analyst forecasts provide a market-based measure of the belief environment in which managers form expectations. Throughout, I interpret optimism or pessimism in analyst forecasts as indicative of the belief distortions surrounding the firm’s decisions.

IBES analysts provide forecasts of “street” earnings per share for each firm’s upcoming fiscal years, typically over horizons of one to four years ahead, as well as a long-term growth (LTG) forecast representing the expected average annual growth in operating earnings over the next three to five years. Released monthly and widely followed, these firm-specific forecasts capture firm-level subjective expectations that can be aggregated to the market level.

For each firm i , I compute expected earnings growth as the forward annual log difference between adjacent horizon level forecasts. Specifically, for $h = 1, 2$, the forecasted annual growth rate $\mathbb{F}_t[\Delta e_{i,t+h}]$ is the log change between the h -year-ahead and $(h - 1)$ -year-ahead level forecasts. For horizons $h = 3, 4, 5$, I interpret the LTG forecast as the expected annualized growth rate over the next three to five years, following De La O and Myers (2021).

Aggregate level. Firm-level forecasts are aggregated to the S&P 500 using market-capitalization value weights (value-weighted averages for long-term growth forecasts). The result proxies subjective expectations of future corporate cash flows at both the firm and aggregate levels.

¹¹Appendix Table A.1 reports robustness checks using managerial earnings forecasts from the quarterly CFO survey. The correlation between analyst and managerial expectations is 0.60 at the one-year horizon.

Discount Rates (Stock Returns) Subjective expectations of stock returns $\mathbb{F}_t[r_{t+h}]$ are drawn from two survey sources, depending on whether the analysis is at the aggregate or firm level.

Aggregate level. For the S&P 500, I use the quarterly *CFO Global Business Outlook Survey* (2001Q4–2023Q4), which reports the mean expected nominal return on the S&P 500 over horizons of one and ten years. Respondents include chief financial officers, vice presidents of finance, and other senior financial executives from a broad set of U.S. firms ranging from small private companies to large publicly listed corporations (approximately 1,600 members as of 2022). For intermediate horizons between one and ten years, I interpolate linearly between the reported forecasts.

Firm level. For individual firms, I use stock price target data from the *IBES* and *Value Line* databases, which report the median forecasted stock price over 12 month and 3 to 5 year horizons, respectively. These forecasts are provided by sell-side equity research analysts who specialize in evaluating firm fundamentals and valuation prospects. I convert these price targets into implied expected total returns as

$$\mathbb{F}_t[r_{i,t+h}] \approx \log\left(\frac{\mathbb{F}_t[P_{i,t+h}]}{P_{i,t}} + \frac{D_{i,t}}{P_{i,t}} \frac{\mathbb{F}_t[D_{i,t+h}]}{D_{i,t}}\right),$$

where $P_{i,t}$ is the firm’s current stock price (CRSP), $D_{i,t}$ is its dividend (Compustat), and the expected dividend growth ratio $\mathbb{F}_t[D_{i,t+h}]/D_{i,t}$ is set to 1.064, the postwar average for U.S. equities (Nagel and Xu, 2022b). I interpret Value Line targets as five-year-ahead forecasts and interpolate intermediate horizons linearly between the IBES one-year and Value Line five-year projections.

Price-Earnings Ratio The current log price-earnings ratio is $pe_{i,t} \equiv \log(P_{i,t}/E_{i,t})$. I construct $\mathbb{F}_t[pe_{i,t+h}]$ using the (Campbell and Shiller, 1988) present-value identity:

$$\mathbb{F}_t[pe_{i,t+h}] = \frac{1}{\rho^h} pe_{i,t} - \frac{1}{\rho^h} \sum_{j=1}^h \rho^{j-1} (c_{pe} + \mathbb{F}_t[\Delta e_{i,t+j}] - \mathbb{F}_t[r_{i,t+j}]),$$

where c_{pe} is a constant and $\rho = \frac{\exp(\overline{pe})}{(1+\exp(\overline{pe}))}$ is the discount factor from the log-linearization. I construct time-series data on actual and expected log price-earnings similarly by using the corresponding aggregate time series data at the S&P 500 level.

Machine Learning Forecasts For each survey forecast, I construct the corresponding machine learning forecast using Long Short-Term Memory (LSTM) neural networks following Bianchi et al. (2022, 2024, 2025). The LSTM model predicts stock returns or earnings growth $y \in \{r, \Delta e\}$

at horizons $h = 1, \dots, 5$ years using a data-rich set of real-time predictors.¹²

$$\mathbb{E}_t[y_{t+h}] = G(\mathcal{X}_t, \beta_t^{TS}; \lambda_t^{TS}), \quad \mathbb{E}_t[y_{i,t+h}] = G(\mathcal{X}_{i,t}, \beta_t^{CS}; \lambda_t^{CS}),$$

where $G(\cdot)$ is the nonlinear mapping learned by the LSTM network. The time-series predictor set $\mathcal{X}_t$ includes real-time macroeconomic, financial, and textual variables (Latent Dirichlet Allocation (LDA)-based news sentiment from the *Wall Street Journal*), as well as survey forecasts. Including the survey forecast allows the machine to combine public information with intangible private information embedded in the survey responses.¹³ The cross-sectional set $\mathcal{X}_{i,t} = \mathcal{X}_t \otimes \mathcal{C}_{i,t}$ augments $\mathcal{X}_t$ with firm characteristics $\mathcal{C}_{i,t}$ including valuation, profitability, size, momentum, volatility, and industry dummies (Chen and Zimmermann, 2022).

The parameters β_t^{TS} and β_t^{CS} are re-estimated dynamically over rolling samples (quarterly for the time-series model and annually for the cross-sectional model) to allow for evolving relationships between predictors and outcomes. In the cross-sectional specification, all firms share the same estimated parameters β_t^{CS} , so differences in predicted outcomes arise only from firm-specific inputs $\mathcal{X}_{i,t}$. Regularization hyperparameters λ_t^{TS} and λ_t^{CS} control model complexity using L_1 and L_2 penalties, dropout layers, early stopping, and ensemble averaging. These techniques prevent overfitting and ensure smooth updates in the presence of structural change or regime shifts. The model’s out-of-sample testing period spans 2005Q1–2023Q4.

I interpret the machine forecast as the ex-ante belief of an agent who processes large amounts of information efficiently under a shrinkage prior. The regularization controlled by λ_t^{TS} and λ_t^{CS} corresponds to Bayesian estimation under a complexity prior, with the forecast as the posterior mean that minimizes expected squared loss. Importantly, the machine must make its forecast using only real-time information, without look-ahead bias. All inputs are restricted to historical, timestamped data that we verify were accessible to market participants at the time, ensuring the predictions reflect only information a firm manager could have used when deciding whether to hire.

3 Evidence of Belief Distortions

Predictability of Survey vs. Machine Forecast Errors To assess whether survey expectations deviate from objective expectations, Figure 1 reports the coefficient β from Coibion and

¹²Appendix OD provides additional details about the machine learning algorithm and the list of predictors.

¹³While the machine uses survey forecasts as one input among many predictive variables, a rational agent would efficiently extract signals from all available public information, including potentially biased survey data, and de-bias them optimally. The machine learning model approximates this process by finding the optimal forecast given an information set that includes survey responses alongside other predictive variables.

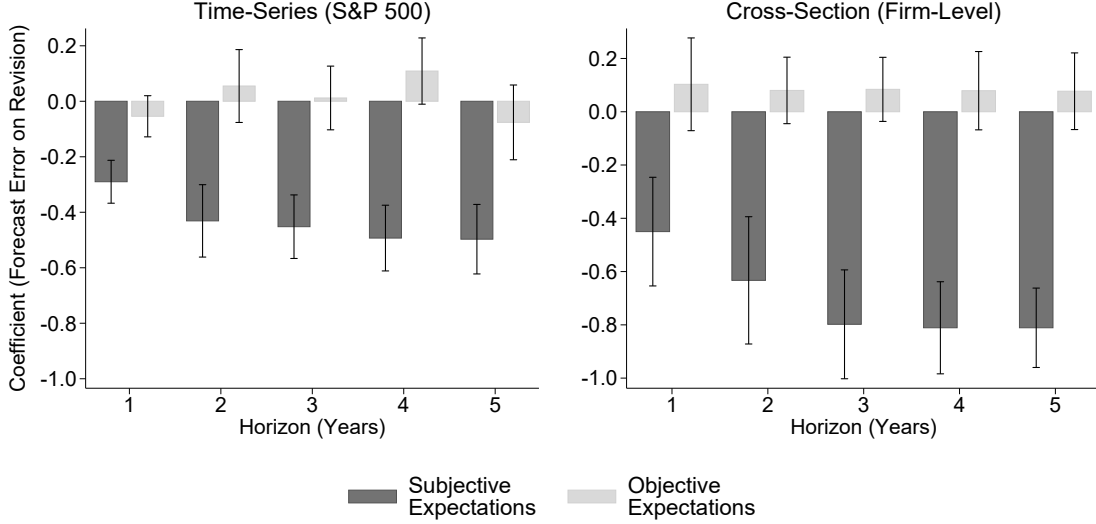
Gorodnichenko (2015) regressions of survey forecast errors on survey forecast revisions at both the aggregate and firm levels:

$$\Delta e_{t+h} - \mathbb{F}_t[\Delta e_{t+h}] = \beta[\mathbb{F}_t[\Delta e_{t+h}] - \mathbb{F}_{t-1}[\Delta e_{t+h}]] + \varepsilon_t \quad (1)$$

$$\Delta e_{i,t+h} - \mathbb{F}_t[\Delta e_{i,t+h}] = \beta[\mathbb{F}_t[\Delta e_{i,t+h}] - \mathbb{F}_{t-1}[\Delta e_{i,t+h}]] + \alpha_i + \alpha_t + \varepsilon_{i,t} \quad (2)$$

where Δe_{t+h} denotes aggregate earnings growth (cash flows) in the time-series regression, and $\Delta e_{i,t+h}$ denotes firm-level earnings growth in the cross-sectional regression, as defined in Section 2. The cross-sectional regressions include firm α_i and time α_t fixed effects, implying that the identifying variation comes from revisions in expectations that are idiosyncratic to each firm.

Figure 1: Predictability of Survey vs. Machine Forecast Errors of Cash Flows



Notes: Figure reports regression coefficients β from regressions of forecast errors on forecast revisions. Left panel: time-series regressions for the S&P 500. Right panel: cross-sectional regressions for a quarterly panel of listed firms with firm and time fixed effects. The forecast target is earnings growth Δe_{t+h} (cash flows). Time-series survey forecasts $\mathbb{F}_t$ come from IBES. Cross-sectional survey forecasts $\mathbb{F}_t$ come from IBES. Machine learning expectations $\mathbb{E}_t$ are generated using a Long Short-Term Memory (LSTM) model trained in real time on macroeconomic, financial, textual, and firm-level data. The sample covers quarterly data from 2005Q1 to 2023Q4. Whiskers show 95% confidence intervals (Newey-West with 4 lags for time-series; two-way clustered by firm and time for cross-section).

In both the time series (left panel) and the firm cross-section (right panel), the coefficients on survey forecast revisions are negative, indicating overreaction; at the firm level they range from -0.433 at one year to -0.818 at five years. Upward revisions are followed by negative forecast errors, so survey respondents respond too strongly to positive earnings news. In contrast, the coefficients on machine forecast revisions are near zero and insignificant at all horizons, as expected for objective expectations, whose forecast errors should be unpredictable.

One might worry that survey forecasts incorporate private information unavailable to the machine, so that apparent distortions reflect rational use of superior information. But if analysts

incorporated their signals efficiently, their revisions would not predict subsequent errors; instead, positive revisions systematically predict negative errors, indicating overreaction. Distortions can also arise even with private information if agents overweight its precision or combine it inefficiently with public data (Bianchi et al., 2022).

Accuracy of Machine Learning vs. Survey Forecasts To assess whether survey respondents misweight relevant information, Figure 2 compares the out-of-sample accuracy of machine and survey forecasts for cash flows e_{t+h} and discount rates r_{t+h} , the two determinants of the value of hiring. I measure relative performance by the ratio $MSE_{\mathbb{E}}/MSE_{\mathbb{F}}$ of machine to survey mean-squared forecast error:

$$\text{Machine MSE} = \frac{1}{T} \sum_{t=1}^T (y_{t+h} - \mathbb{E}_t[y_{t+h}])^2, \quad \text{Survey MSE} = \frac{1}{T} \sum_{t=1}^T (y_{t+h} - \mathbb{F}_t[y_{t+h}])^2$$

where y_{t+h} denotes either cash flows or discount rates, and T is the length of the out-of-sample testing period, which spans 2005Q1 to 2023Q4. The cross-sectional forecasts are evaluated similarly based on the mean-squared-forecast-error across all firms over the testing period.

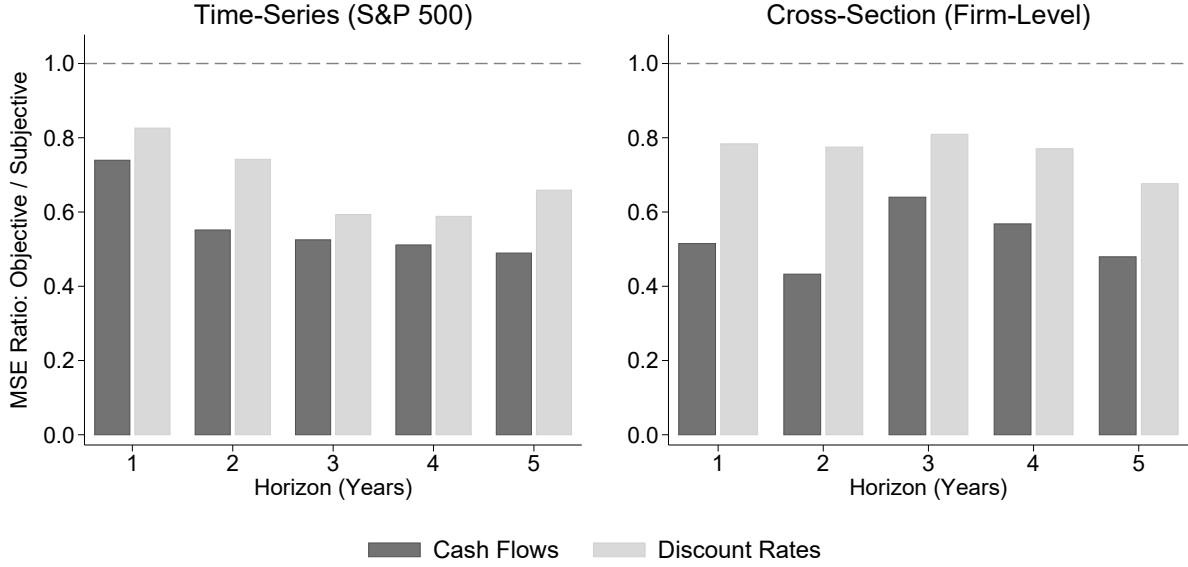
Figure 2 shows that machine learning forecasts consistently outperform survey forecasts across all variables and horizons, with MSE ratios well below one. The performance gap widens with forecast horizon, indicating that belief distortions are larger at longer horizons. The machine outperforms the survey for both time-series and cross-sectional forecasts, suggesting that belief distortions affect not only aggregate expectations but also the cross-sectional dispersion of beliefs across firms. If survey respondents were efficient in forming their beliefs, their forecasts would have performed at least on par with the machine.¹⁴ The superior performance of the machine also highlights its ability to process a large amount of real-time data efficiently and objectively, supporting its use as a reliable benchmark of undistorted beliefs.

Because the machine conditions on the survey forecasts themselves, it already incorporates any private information they embed while correcting for its inefficient use. That the machine still outperforms therefore reflects systematic biases in belief formation, not superior private information.

Hiring Outcomes and Belief Distortions Given the evidence of overreaction in subjective cash flow forecasts, this section examines the relationship between the vacancy filling rate and the

¹⁴The magnitude of the MSE ratio is more consistent with truly subjective beliefs rather than objective but risk-neutral beliefs. Estimates of risk premia typically range 5-10% annually (Adam et al., 2021), which is insufficient to explain the 15-30% deterioration in MSE ratios observed in Figure 2. The magnitude and persistence of forecast errors across horizons instead point to behavioral biases rather than risk compensation.

Figure 2: Accuracy of Machine Learning vs. Survey Forecasts



Notes: Figure plots MSE_E/MSE_F , the ratio of mean squared forecast errors between machine learning and survey forecasts. Lower values indicate greater accuracy of the machine learning forecast. MSE_E and MSE_F refer to out-of-sample forecast errors from machine and survey forecasts, respectively. The out-of-sample testing period is 2005Q1–2023Q4. Left panel: aggregate S&P 500 time-series results. Right panel: cross-sectional forecasts across listed firms. Dark bars show cash flows (earnings growth); light bars show discount rates (stock returns). The forecast target y_{t+h} is the present value of discount rates $r_{t,t+h}$ and cash flows $e_{t,t+h}$, as defined in equation (17). Time-series survey forecasts F_t come from the CFO survey (discount rates) and IBES (cash flows). Cross-sectional survey forecasts F_t come from IBES (discount rates and cash flows). Time-series and cross-sectional machine learning expectations E_t are generated using a Long Short-Term Memory (LSTM) model trained in real time on macroeconomic, financial, and textual data.

belief distortion in subjective five-year cash flow expectations. In search-and-matching models of the labor market, the vacancy filling rate reflects the marginal value of job creation. Hiring creates a match with expected duration $1/\delta$ years given a job separation rate δ . A five-year forecast horizon captures the lifetime of a typical match: median job tenure ranged from 3.4 to 4.6 years over 1983–2023 (Bureau of Labor Statistics, 2024), and the right-skewed tenure distribution implies an expected duration above the median.

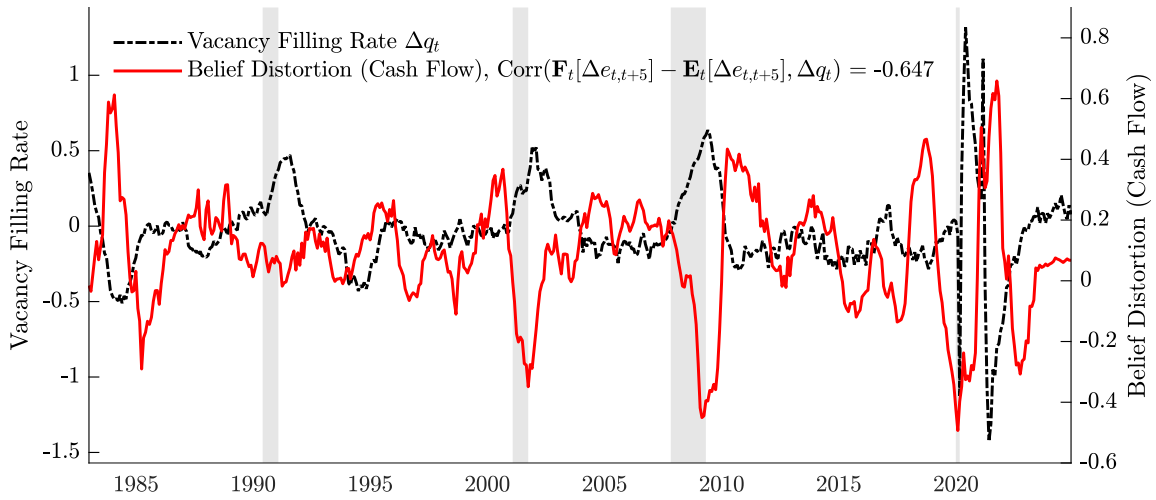
Figure 3 plots the growth rate of the U.S. vacancy filling rate against belief distortions in cash flow growth expectations, measured as the gap between subjective and objective five-year forecasts of annualized S&P 500 earnings growth ($F_t[\Delta e_{t,t+5}] - E_t[\Delta e_{t,t+5}]$). The subjective forecasts come from survey data, while the objective benchmark uses machine learning predictions.

The figure shows that belief distortions exhibit strong cyclical patterns that closely track labor market fluctuations, with a negative correlation of -0.647. During expansions, positive belief distortions emerge as subjective forecasts become overly optimistic relative to the objective benchmark. These periods of optimism coincide with declines in the vacancy filling rate. The negative correlation reflects a congestion effect: when firms become optimistic, they simultaneously surge in vacancy posting, which tightens the labor market and makes it harder for any

given vacancy to be filled. Conversely, when belief distortions turn negative and firms scale back vacancy posting, the vacancy filling rate rises as the market becomes less congested.

This pattern is particularly evident during the large positive belief distortions of the pre-2008 period, followed by a dramatic reversal to negative distortions that coincided with a sharp contraction in vacancy filling rate growth and the 2008–2009 NBER-dated recession. The tight co-movement between these expectational errors in cash flow forecasts and vacancy filling rate growth suggests that distortions in firms’ earnings expectations are a powerful driver of labor market fluctuations, operating independently of changes in discount rates.

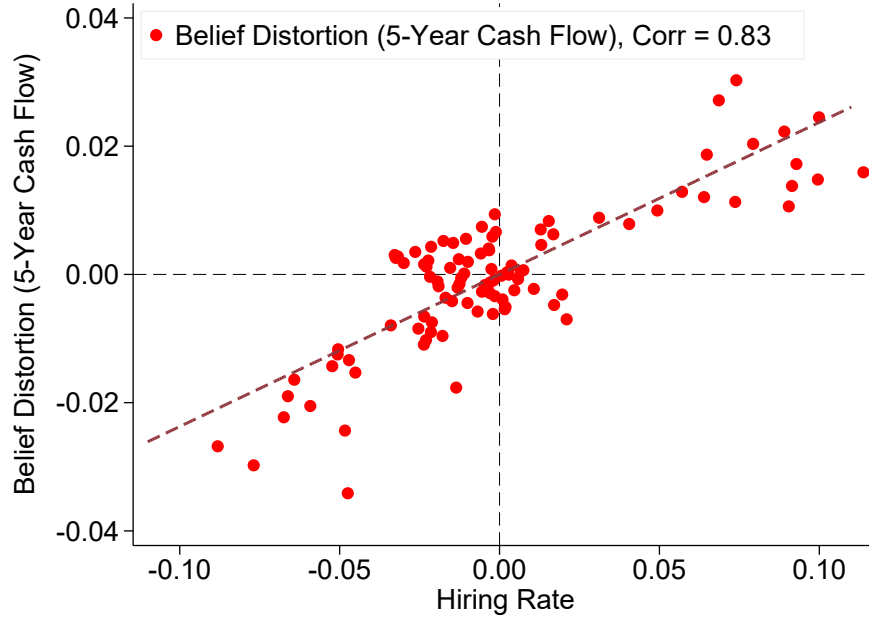
Figure 3: Vacancy Filling Rate and Belief Distortions in Subjective Cash Flows



Notes: Figure plots the annual log growth of the U.S. vacancy filling rate Δq_t (left axis) against the belief distortion in cash flows, which is measured as expectational errors $\mathbb{F}_t[\Delta e_{t,t+5}] - \mathbb{E}_t[\Delta e_{t,t+5}]$ in five-year forecasts of annualized S&P 500 earnings growth (right axis). Survey expectations $\mathbb{F}_t[\Delta e_{t+5}]$: IBES median analyst forecasts for the next four fiscal years and long-term growth (LTG). Objective expectations $\mathbb{E}_t[\Delta e_{t+5}]$: Machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. The vacancy filling rate, defined as total hires divided by total vacancies, measures the rate at which posted job openings are filled. Sample is quarterly from 2005Q1 to 2023Q4, the period over which the machine forecast is available out of sample. Gray shaded areas indicate NBER recessions.

Figure 4 shows the analogous cross-sectional relationship, plotting firm-level hiring rates against belief distortions in cash flow expectations across listed firms. The scatter shows a clear positive relationship (correlation 0.830): firms whose analyst coverage is excessively optimistic relative to the machine about five-year cash flow growth hire systematically more. Each point is a percentile bin, and the positive slope confirms that belief distortions in expected cash flows translate directly into hiring differences across firms. Intuitively, the positive firm-level correlation in the hiring rate complements the negative aggregate relationship for the vacancy filling rate. When firms become more optimistic, they expand employment and raise hiring rates, but collectively this tightens the labor market and lowers the probability that each vacancy is filled.

Figure 4: Cross-Sectional Hiring Rates and Belief Distortions



Notes: Figure plots the relationship between hiring rates and cash flow expectations across listed firms. y -axis reports the corresponding cross-sectionally demeaned log hiring rate, $hl_{i,t}$. x -axis reports the cross-sectionally demeaned belief distortion measured as expectational errors $\mathbb{F}_t[\Delta e_{t,t+5}] - \mathbb{E}_t[\Delta e_{t,t+5}]$ in five-year forecasts of annualized S&P 500 earnings growth. Subjective expectations $\mathbb{F}_t$: IBES forecasts. Objective expectations $\mathbb{E}_t$: Machine learning (LSTM) forecasts. The hiring rate, defined as total hires divided by existing employment, captures firm-level hiring behavior. Each dot is a bin scatter representing one percentile of the pooled distribution across all observations in the sample. The sample is quarterly from 2005Q1 to 2023Q4.

4 Theoretical Framework

The reduced-form link between fluctuations in the vacancy filling rate and belief distortions about cash flows motivates a structural interpretation. This section develops a search and matching model of the labor market in which firms' expectations about future cash flows and discount rates may be distorted, leading to fluctuations in vacancy filling rates and unemployment. The structure here is the reduced-form framework that the variance decompositions of Sections 5 and 6 rest on; its full general-equilibrium counterpart, with Epstein-Zin households, limited asset-market participation, and a diagnostic growth belief, is developed and calibrated in Section 7. The model builds on the Diamond (1982), Mortensen (1982), and Pissarides (2009) framework but departs from the standard rational expectations assumption, allowing firms' hiring decisions to be influenced by biased subjective beliefs.

Environment Consider a discrete time economy populated by a representative household and a mass of firms I , normalized to one, that hires workers in a frictional labor market. The firm uses labor as a single input to production. The household's population is normalized to one and

has a continuum of members, where a fraction L_t are employed and the rest are unemployed $U_t = 1 - L_t$. The household's intertemporal consumption decision gives rise to a stochastic discount factor M_{t+1} .

Labor Market Each period, each firm posts job vacancies at a cost $\kappa > 0$ to maximize its cum-dividend value of equity. Employment $L_{i,t}$ reflects the number of workers employed in firm i at the beginning of period t before any separations or new hires.¹⁵ During the period, a fraction $\delta_{i,t}$ of employed workers separate, while the firm posts vacancies $V_{i,t}$ to search from a pool of unemployed workers U_t . Let $L_t = \sum_{i \in I} L_{i,t}$ and $V_t = \sum_{i \in I} V_{i,t}$ denote the aggregate number of employed workers and vacancies posted by all firms. Matches are formed at the end of period t according to a Cobb-Douglas matching function $\mathcal{M}(U_t, V_t)$:

$$\mathcal{M}(U_t, V_t) = \chi U_t^\alpha V_t^{1-\alpha} \quad (3)$$

where χ is the matching efficiency parameter, and $\alpha \in (0, 1)$ governs the elasticity of matches with respect to unemployment. The probability that a firm fills a posted vacancy, the vacancy filling rate, is then given by:

$$q_t = \frac{\mathcal{M}(U_t, V_t)}{V_t} = \chi \left(\frac{U_t}{V_t} \right)^\alpha = \chi \theta_t^{-\alpha} \quad (4)$$

and the job-finding rate is given by $f_t \equiv \mathcal{M}(U_t, V_t)/U_t$. These new hires enter employment at the start of period $t + 1$, so employment $L_{i,t}$ and unemployment $U_t = 1 - L_t$ evolve according to the law of motion:

$$L_{i,t+1} = (1 - \delta_{i,t})L_{i,t} + q_t V_{i,t} \quad (5)$$

Aggregate unemployment $U_t = 1 - L_t$ evolves according to:

$$U_{t+1} = \delta_t(1 - U_t) + (1 - f_t)U_t \quad (6)$$

where $\theta_t = V_t/U_t$ denotes labor market tightness, defined as the vacancy-to-unemployment ratio.

Firm's Technology and Cash Flow Each firm i uses labor $L_{i,t}$ to produce output $Y_{i,t}$ via a Cobb-Douglas production function under constant returns to scale:

$$Y_{i,t} = A_{i,t} L_{i,t} \quad (7)$$

¹⁵I adopt an end-of-period matching convention following Petrosky-Nadeau et al. (2018). See Hansen et al. (2005) and Kogan and Papanikolaou (2012) for similar conventions applied for the q theory of investment.

where $A_{i,t}$ is firm-level productivity and $L_{i,t}$ is the level of employment. The firm pays wages $W_{i,t}$, incurs hiring costs $\kappa V_{i,t}$, and generates cash flows defined as the firm's period earnings:

$$E_{i,t} = Y_{i,t} - W_{i,t}L_{i,t} - \kappa V_{i,t} \quad (8)$$

Earnings represent the net flow profits from operating the firm, which is the output net of the wage bill and vacancy posting costs. I assume that the household owns the equity of the firm and the firm pays out all of its earnings $E_{i,t}$ as dividends (Petrosky-Nadeau et al., 2018). I also assume that the firm's manager has access to complete markets so that the return to hiring equals the stock market return in equilibrium (Cochrane, 1991).

Firm's Problem The firm chooses vacancy postings $V_{i,t}$ to maximize the present discounted value of future cash flows. The firm's value function $\mathcal{V}$ satisfies the Bellman equation:

$$\mathcal{V}(A_{i,t}, L_{i,t}) = \max_{V_{i,t}, L_{i,t+1}} \{E_{i,t} + \mathbb{E}_t[M_{t+1}\mathcal{V}(A_{i,t+1}, L_{i,t+1})]\} \quad (9)$$

subject to the employment accumulation equation (5). $\mathbb{E}_t[\cdot]$ is the firm's subjective expectations conditional on information available at the beginning of period t .¹⁶ These beliefs may depart from objective expectations $\mathbb{E}_t[\cdot]$, where the nature and magnitude of the deviation will be disciplined using survey data. M_{t+1} is the stochastic discount factor that prices the firm's cash flows. The firm does not observe this discount factor directly and needs to form expectations about it by forecasting the household's marginal utility of consumption (Venkateswaran, 2014).

Hiring Condition Under search frictions, hiring is forward-looking investment. The firm's optimal hiring decision equates the marginal cost of posting a vacancy with the expected discounted marginal value of employment:

$$\underbrace{\frac{\kappa}{q_t}}_{\text{Cost of hiring}} = \mathbb{E}_t \left[\underbrace{M_{t+1} \frac{\partial \mathcal{V}(A_{i,t+1}, L_{i,t+1})}{\partial L_{i,t+1}}}_{\text{Expected discounted value of hiring}} \right] \quad (10)$$

The left side represents the expected cost of hiring an additional worker while accounting for the probability q_t that a posted vacancy will be filled. The right side captures the expected discounted value of the marginal worker, incorporating both the firm's subjective beliefs about the future state and the discount rate for valuing risky cash flows.¹⁷ Subjective distortions in beliefs can

¹⁶I use the term "firm's beliefs" as shorthand to refer to the expectations held by decision makers within firms (Coibion et al., 2018; Candia et al., 2020).

¹⁷The hiring equation is the labor market analogue of the optimality condition for physical capital in the q theory of investment (Hayashi, 1982), where the upfront cost of hiring κ/q_t is analogous to Tobin's marginal q and the separation rate δ_{t+1} is analogous to the depreciation rate (Borovičková and Borovička, 2017). See Lettau and Ludvigson (2002) and Kogan and Papanikolaou (2012) for a similar log-linearization applied for the q theory of physical capital investment.

thus shift the perceived value of hiring through $\mathbb{E}_t[\cdot]$ and affect equilibrium vacancy filling rates, which in turn affect unemployment through its law of motion in equation (6). Assuming constant returns to scale, the marginal value of hiring equals its average value:

$$\frac{\partial \mathcal{V}(A_{i,t+1}, L_{i,t+1})}{\partial L_{i,t+1}} = \frac{\mathcal{V}(A_{i,t+1}, L_{i,t+1})}{L_{i,t+1}} \quad (11)$$

Define the firm's ex-dividend market value as $P_{i,t} \equiv \mathbb{E}_t[M_{t+1} \mathcal{V}(A_{i,t+1}, L_{i,t+1})]$ to derive a direct link between the vacancy filling rate and the firm's market value per worker:

$$\frac{\kappa}{q_t} = \frac{P_{i,t}}{L_{i,t+1}} \quad (12)$$

where employment $L_{i,t+1}$ is determined at the end of date t under the timing convention from equation (5). Take logarithms, rearrange terms, and expand the price-employment ratio $P_{i,t}/L_{i,t+1}$:

$$\log q_t = \log \kappa - \log \left(\frac{P_{i,t}}{E_{i,t}} \right) - \log \left(\frac{E_{i,t}}{L_{i,t+1}} \right) \quad (13)$$

Define log price-earnings $pe_{i,t} = \log(P_{i,t}/E_{i,t})$ and earnings-employment $el_{i,t} = \log(E_{i,t}/L_{i,t+1})$:

$$\log q_t = \log \kappa - pe_{i,t} - el_{i,t} \quad (14)$$

Log-linear Approximation of Price-Earnings Ratio To decompose the vacancy filling rate into economically meaningful components, I apply the Campbell and Shiller (1988) present value identity to the price-earnings ratio. Log-linearize the price-earnings ratio $pe_{i,t} \equiv \log(P_{i,t}/E_{i,t})$ around its long-run mean $\overline{pe}$ to obtain the approximate relationship:

$$pe_{i,t} = c_{pe} - r_{i,t+1} + \Delta e_{i,t+1} + \rho pe_{i,t+1} \quad (15)$$

where c_{pe} is a linearization constant, $\rho = \exp(\overline{pe})/(1 + \exp(\overline{pe}))$ is the time discount factor from the log-linearization, $r_{i,t+1} = \log((P_{i,t+1} + E_{i,t+1})/P_{i,t})$ is the stock return assuming that the firm pays out its earnings as dividends, and $\Delta e_{i,t+1}$ denotes earnings growth.¹⁸ This equation is an accounting identity that links current valuation ratios to future cash flows and discount rates. Substituting recursively for the next h periods yields the present value identity:

$$pe_{i,t} = \sum_{j=1}^h \rho^{j-1} c_{pe} - \sum_{j=1}^h \rho^{j-1} r_{i,t+j} + \sum_{j=1}^h \rho^{j-1} \Delta e_{i,t+j} + \rho^h pe_{i,t+h} \quad (16)$$

¹⁸This identity also holds approximately when dividends differ from earnings. Following Lintner (1956), dividends can be approximated as a stable fraction of earnings (with a long-run payout ratio near 50%). The resulting payout ratio term $(1 - \rho)de_{t+1}$ becomes negligible after log-linearization since $1 - \rho \approx 0.02$, allowing this term to be absorbed into the constant c_{pe} (De La O et al., 2024). See Appendix Section OB for a derivation.

Decomposition of Vacancy Filling Rate Substitute log-linearized price-earnings (16) into the hiring equation (14) to obtain a decomposition of the vacancy filling rate q_t :

$$\log q_t = c_q + \underbrace{\sum_{j=1}^h \rho^{j-1} r_{i,t+j}}_{\equiv r_{i,t,t+h}} - \underbrace{\left[el_{i,t} + \sum_{j=1}^h \rho^{j-1} \Delta e_{i,t+j} \right]}_{\equiv e_{i,t,t+h}} - \underbrace{\rho^h pe_{i,t+h}}_{\equiv pe_{i,t,t+h}} \quad (17)$$

where $c_q \equiv \log \kappa - \frac{c_{pe}(1-\rho^h)}{1-\rho}$ is a constant. The vacancy filling rate has been decomposed into three forward-looking components: the present value of future discount rates $r_{i,t,t+h} \equiv \sum_{j=1}^h \rho^{j-1} r_{i,t+j}$, cash flows $e_{i,t,t+h} \equiv el_{i,t} + \sum_{j=1}^h \rho^{j-1} \Delta e_{i,t+j}$, and price-earnings ratio $pe_{i,t,t+h} \equiv \rho^h pe_{i,t+h}$. The cash flow component consists of the current earnings-employment ratio el_t , which captures short-term fluctuations in cash flows, and $j = 1, \dots, h$ period ahead earnings growth Δe_{t+j} , which captures news about future cash flows. The persistence of labor matches enters through the continuation value term $\rho^h \mathbb{F}_t[pe_{t+h}]$. In the firm's Bellman equation, match persistence determines how much of the job's value extends beyond the h -year horizon. A lower separation rate (higher match persistence) raises the continuation value, increasing the contribution of the terminal value component in the decomposition.

To decompose time-series variation in the vacancy filling rate q_t , the right-hand side of equation (17) can be aggregated across firms:

$$\underbrace{\log q_t}_{\text{Vacancy Filling Rate}} = c_q + \underbrace{\mathbb{F}_t[r_{t,t+h}]}_{\text{Discount Rate}} - \underbrace{\mathbb{F}_t[e_{t,t+h}]}_{\text{Cash Flow}} - \underbrace{\mathbb{F}_t[pe_{t,t+h}]}_{\text{Future Price-Earnings}} \quad (18)$$

where $x_t = \sum_{i \in I} w_{i,t} x_{i,t}$ aggregates firm-level variable $x_{i,t}$ using employment weights $w_{i,t} = L_{i,t} / \sum_{j \in I} L_{j,t}$ for $x \in \{r, e, pe\}$.¹⁹ Intuitively, the vacancy filling rate falls when firms expect high future cash flows (making hiring valuable) or low future discount rates (making future profits more valuable today). The Campbell-Shiller formula allows us to separate these two channels by recursively decomposing the price-earnings ratio into expected returns and earnings growth.

Since equation (17) holds both ex-ante and ex-post, it can be evaluated under either subjective or objective expectations. The *subjective decomposition* replaces ex-post realizations of future outcomes with their ex-ante subjective expectation $\mathbb{F}_t[\cdot]$:

$$\log q_t = c_q + \mathbb{F}_t[r_{t,t+h}] - \mathbb{F}_t[e_{t,t+h}] - \mathbb{F}_t[pe_{t,t+h}] \quad (19)$$

¹⁹The aggregate decomposition follows from the firm-level decomposition by summing across individual firms' hiring conditions under two assumptions: (i) vacancy posting costs are linear in the number of vacancies, and (ii) the labor market for matches is competitive, so that all firms face a common vacancy filling rate q_t . Under these conditions, the aggregate hiring equation is obtained by weighting firm-level equations by employment, which ensures that the decomposition holds exactly at the aggregate level. For the cross-sectional analysis, the same logic applies to portfolios of firms.

The equation implies that the vacancy filling rate is high when firms subjectively expect future returns to be high, expected cash flows to be low, or both. Alternatively, the *objective decomposition* replaces each subjective expectation $\mathbb{F}_t[\cdot]$ with its objective expectation $\mathbb{E}_t[\cdot]$:

$$\log q_t = c_q + \mathbb{E}_t[r_{t,t+h}] - \mathbb{E}_t[e_{t,t+h}] - \mathbb{E}_t[pe_{t,t+h}] \quad (20)$$

Comparing these decompositions can quantify how belief distortions affect the vacancy filling rate. The econometrician can estimate the variance decomposition using predictive regressions of each expected outcome on the current vacancy filling rate. For the subjective decomposition, demean each variable in equation (19), multiply both sides by the current log vacancy filling rate $\log q_t$, and take the sample average:

$$Var[\log q_t] = Cov[\mathbb{F}_t[r_{t,t+h}], \log q_t] - Cov[\mathbb{F}_t[e_{t,t+h}], \log q_t] - Cov[\mathbb{F}_t[pe_{t,t+h}], \log q_t] \quad (21)$$

where $Var[\cdot]$ and $Cov[\cdot]$ are sample variances and covariances based on data observed over a historical sample. Finally, divide both sides by $Var[\log q_t]$ to decompose its variance:

$$1 = \underbrace{\frac{Cov[\mathbb{F}_t[r_{t,t+h}], \log q_t]}{Var[\log q_t]}}_{\text{Discount Rate News}} - \underbrace{\frac{Cov[\mathbb{F}_t[e_{t,t+h}], \log q_t]}{Var[\log q_t]}}_{\text{Cash Flow News}} - \underbrace{\frac{Cov[\mathbb{F}_t[pe_{t,t+h}], \log q_t]}{Var[\log q_t]}}_{\text{Future Price-Earnings News}} \quad (22)$$

The left-hand side represents the full variability in vacancy filling rates, hence is equal to one. Each term on the right reflects the share explained by subjective expectations of discount rates, cash flows, or future price-earnings ratios. Under stationarity, the econometrician can estimate these shares using the OLS coefficients from regressing $\mathbb{F}_t[r_{t,t+h}]$, $\mathbb{F}_t[e_{t,t+h}]$, and $\mathbb{F}_t[pe_{t,t+h}]$ on the current log vacancy filling rate $\log q_t$, respectively. Finally, the decomposition under objective expectations can be estimated similarly based on equation (22) by replacing the subjective expectation $\mathbb{F}_t[\cdot]$ with its objective counterpart $\mathbb{E}_t[\cdot]$.

Comparing the decompositions implied by subjective and objective expectations can highlight the role of belief distortions, which I define as the gap between the survey and machine forecasts: $\mathbb{F}_t - \mathbb{E}_t$. This comparison allows us to assess the role of belief distortions in explaining labor market dynamics and determine whether firms systematically mis-perceive economic conditions when making hiring decisions. Although the variance decomposition does not necessarily capture causal relationships, it has the advantage of not requiring the researcher to take a stand on the deep determinants of vacancy filling rates because the evolution of discount rates and cash flows summarize the combined effects of these deep determinants.

5 Time-Series Decomposition of the Vacancy Filling Rate

The evidence of overreaction in survey expectations and the superior forecasting performance of machine learning suggests the presence of distortions in subjective expectations. This section quantifies how those distortions affect hiring behavior by estimating the contributions of discount rate and cash flow expectations to fluctuations in the aggregate vacancy filling rate. I focus on the vacancy filling rate because it is the matching-function counterpart of the job-finding rate, which Shimer (2005, 2012) show accounts for about three-quarters of unemployment fluctuations; distortions that move the vacancy filling rate therefore move the dominant margin of unemployment volatility.

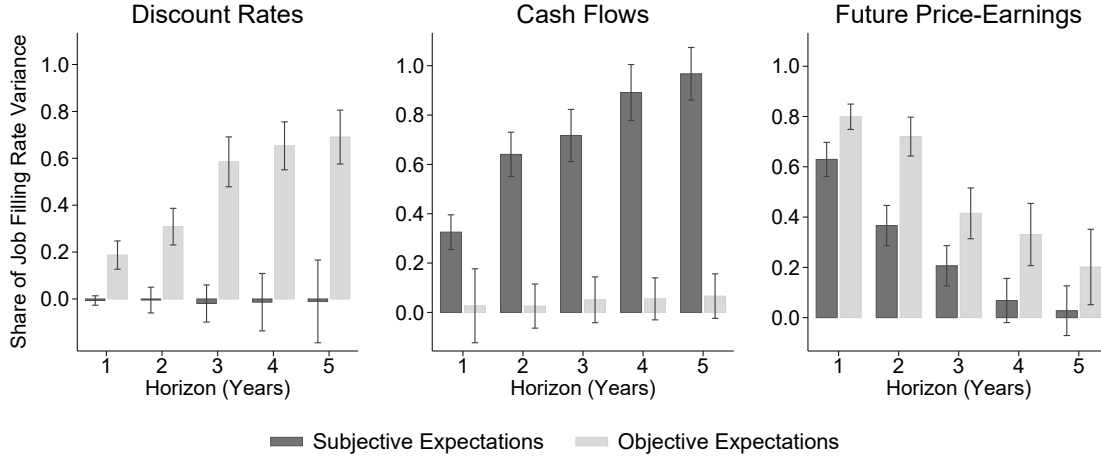
Objective Expectations Figure 5 presents the variance decomposition of the vacancy filling rate. The figure shows that, under objective beliefs, discount rate news is the dominant driver of variation in vacancy filling rates. At the five-year horizon, discount rates explain 69.1% of the variation in vacancy filling rates, while cash flow news accounts for 6.6%.²⁰ Consistent with the predictions of the search and matching model, higher vacancy filling rates are associated with higher discount rates and lower expected cash flows because a lower value of hiring leads firms to post fewer vacancies, reducing tightness and raising q_t . The contribution from terminal price-earnings ratios still remains sizable at the five-year horizon, accounting for 20.1%. The three components sum to 95.8% at the five-year horizon, close to 100%, indicating the decomposition is accurate even though it is estimated freely without imposing this constraint.

Under objective expectations, these findings echo rational search-and-matching models with time-varying risk premia: discount rate news dominates and grows more important at longer horizons, as in models that treat hiring as a risky investment with long-duration returns (Hall, 2017; Kehoe et al., 2023). The small objective cash flow component reflects the unemployment volatility puzzle: Shimer (2005) showed that without time-varying discount rates, productivity shocks alone cannot generate enough unemployment volatility.

Subjective Expectations On the other hand, Figure 5 reveals a strikingly different result under subjective expectations. At the five-year horizon, subjective cash flow news explains 96.7% of the variation in vacancy filling rates, while subjective discount rate news accounts for only -1.0%. Under subjective beliefs, hiring is highly sensitive to cash flow news. Since vacancy filling rates are countercyclical (declining during expansions as labor markets tighten), the positive

²⁰Table OA.1 reports more detailed statistics on the variance decomposition.

Figure 5: Time-Series Decomposition of the Vacancy Filling Rate



Notes: Figure illustrates the discount rate, cash flow, and future price-earnings components from the time-series decomposition of the U.S. aggregate vacancy filling rate. Light bars show contributions under objective expectations; dark bars show contributions under subjective expectations. Subjective expectations $\mathbb{F}_t$ are constructed from CFO survey forecasts (discount rates) and IBES analyst forecasts (cash flows). Objective expectations $\mathbb{E}_t$ are based on machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. x -axis denotes the forecast horizon h . The sample is quarterly from 2005Q1 to 2023Q4. Each bar shows Newey-West 95% confidence intervals with lags = 4.

cash flow coefficient indicates that firms expect high cash flows during periods when q_t is low, consistent with optimism during economic booms. The negative contribution on the discount rate component indicates that survey respondents predict lower future returns during recessions, contrary to what a rational forecast would imply.

The terminal price-earnings ratio's contribution falls with horizon and is negligible by year five (2.8%), compared with 20.1% under objective expectations. This implies that subjective beliefs place larger weight on the firm's near-term cash flows relative to its long-run valuations. Finally, the three components sum to 98.5% at the five-year horizon, showing that survey expectations are internally consistent and the model's approximation is reasonably accurate, with any remaining gap likely attributable to measurement error in the survey data (e.g., Ma et al., 2020).²¹

Compared to the objective benchmark, the implied overreaction to cash flow news is substantial. Low vacancy filling rates during expansions are associated with a significant disappointment in future cash flows. Defining the belief distortion as the difference between subjective and objective expectations $\mathbb{F}_t - \mathbb{E}_t$, the estimates imply that, at the five-year horizon, $96.7\% - 6.6\% = 90.1\%$ of variation in vacancy filling rates can be attributed to the fact that the vacancy filling rate predicts distortions in cash flow expectations with a significant positive relationship (Table OA.2). These distortions capture inefficiencies or behavioral biases in survey respondents' subjective

²¹This residual term captures any model misspecification or approximation errors in the decomposition, including potential misspecification of the stochastic discount factor or approximation errors in the Campbell-Shiller decomposition under subjective beliefs.

beliefs that the machine learning model could have identified ex-ante.

These findings are inconsistent with measurement error or analyst conflicts of interest: random noise would bias results toward zero, and conflicts of interest add bias or noise but not systematic overreaction. The strong predictive power of belief distortions, and the fact that revisions predict subsequent errors, indicates systematic overreaction, not noise.

Sources of Vacancy Filling Rate Variation While Figure 1 has shown subjective discount rate forecasts to exhibit overreaction, their contribution to the variance decomposition of vacancy filling rates remains small in Figure 5. This can be reconciled by the fact that subjective discount rate expectations display relatively little time-series variation, so even overreactive revisions have limited impact on hiring decisions. In contrast, subjective cash flow expectations vary much more over time, making them the primary driver of belief-driven fluctuations in hiring.²²

Appendix B.2 plots counterfactual paths for vacancy filling rate growth driven by each component alone, and confirms that the subjective cash flow component tracks the realized vacancy filling rate far more closely than the objective discount rate component.

Discussion Although the decomposition is not necessarily causal, it identifies the sources of variation in the vacancy filling rate. A large subjective cash flow estimate means that whatever shocks drive the vacancy filling rate load more on subjective cash flow expectations than on discount rates, whereas under objective expectations firms read the same shocks as news about risk compensation. This points to belief distortions as a key source of fluctuations: by overreacting to perceived cash flow changes, firms cut hiring and vacancies too sharply in downturns, amplifying unemployment volatility beyond rational models.

The large contribution of subjective cash flows in shaping hiring decisions is consistent with models that introduce nonrational expectations about earnings growth to account for fluctuations in asset prices (Bordalo et al., 2024a; Bianchi et al., 2024) and the business cycle (Bordalo et al., 2024b). This parallel implies that the belief distortions known to influence asset valuations can also extend to real economic behavior through the labor market.

On the other hand, the small and negative contribution of subjective discount rates (although not statistically different from zero) is consistent with existing survey evidence showing that subjective return expectations are acyclical (Nagel and Xu, 2022b) or even procyclical (Greenwood and Shleifer, 2014; Adam et al., 2016), contrary to the countercyclical discount rate variation

²²Figure A.1 illustrates this point visually by showing that subjective expectations exhibit strong cyclical-ity in cash flow forecasts and muted responses in discount rate forecasts compared to their machine learning counterparts.

implied by rational models (Cochrane, 2017). In standard asset pricing models, discount rates reflect the firm’s market-based cost of capital, such as the weighted average cost of debt and cost of equity (WACC). In contrast, survey evidence suggests that CFOs likely rely on internal discount rates that are persistent and often unresponsive to market conditions, even when firms are not financially constrained (Gormsen and Huber, 2025). My findings extend this evidence to labor markets, where hiring decisions appear similarly detached from subjective beliefs about risk premia or financial constraints.

6 Cross-Sectional Decomposition of the Hiring Rate

To analyze the sources of dispersion in hiring across firms, I implement a cross-sectional decomposition of the log hiring rate based on the same theoretical framework developed for the time-series decomposition. The log hiring rate for each firm can be constructed using the employment accumulation equation:

$$hl_{i,t} = \log \left(\frac{q_t V_{i,t}}{L_{i,t}} \right) = \log \left(\frac{L_{i,t+1}}{L_{i,t}} - (1 - \delta_{i,t}) \right) \quad (23)$$

where $L_{i,t}$ uses data from Compustat number of employees (EMP) and $\delta_{i,t}$ uses JOLTS industry-level job separation rate. The hiring rate captures the fraction of new hires per existing employee, conditional on vacancies being filled at rate q_t . This demeaned hiring rate can be decomposed into three components:²³

$$\begin{aligned} \tilde{hl}_{i,t} = & - \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[\tilde{r}_{i,t+j}]}_{\text{Discount Rate} \equiv \mathbb{F}_t[\tilde{r}_{i,t,t+h}]} + \underbrace{\left[\tilde{el}_{i,t} + \sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[\Delta \tilde{e}_{i,t+j}] \right]}_{\text{Cash Flow} \equiv \mathbb{F}_t[\tilde{e}_{i,t,t+h}]} + \underbrace{\rho^h \mathbb{F}_t[\tilde{pe}_{i,t+h}]}_{\text{Future Price-Earnings} \equiv \mathbb{F}_t[\tilde{pe}_{i,t,t+h}]} \end{aligned} \quad (24)$$

where $\rho = \exp(\overline{pe}) / (1 + \exp(\overline{pe}))$ is the time discount factor from the log-linearization.

The first term represents cross-sectional dispersion in expected returns, which affect the discount rate at which future expected cash flows are converted to present value. The second term captures dispersion in the current earnings per worker, $\tilde{el}_{i,t}$, and the sum of expected earnings growth over the forecast horizon h . The third term is the dispersion in expected future price-earnings ratios, which is a terminal value that captures longer-run influences not already captured in discount rates and expected cash flows by horizon h . All expectations are formed

²³See Appendix Section OB.2 for a derivation. Note that the signs on each component are reversed compared to the vacancy filling rate decomposition in equation (22). The flipped sign reflects the distinction between how easy it is to fill vacancies (vacancy filling rate) versus how much hiring actually occurs (hiring rate). Good economic conditions make vacancies harder to fill but increase overall hiring activity.

using either survey (subjective expectation) or machine learning forecasts (objective expectation benchmark). To isolate cross-sectional variation, I demean each variable across firms indexed by I , defining $\tilde{hl}_{i,t} = hl_{i,t} - \frac{1}{I} \sum_{j \in I} hl_{j,t}$, so that the decomposition isolates the extent to which deviations from the average hiring rate can be traced to each component.

Under stationarity, the econometrician can estimate these shares using OLS coefficients from regressing $\mathbb{F}_t[\tilde{r}_{i,t,t+h}]$, $\mathbb{F}_t[\tilde{e}_{i,t,t+h}]$, and $\mathbb{F}_t[\tilde{pe}_{i,t,t+h}]$ on the current log hiring rate $\tilde{hl}_{i,t}$, respectively. Specifically, demean each variable in equation (24), multiply both sides by the current log hiring rate $\tilde{hl}_{i,t}$, and take the cross-sectional average:

$$Var[\tilde{hl}_{i,t}] = -Cov[\mathbb{F}_t[\tilde{r}_{i,t,t+h}], \tilde{hl}_{i,t}] + Cov[\mathbb{F}_t[\tilde{e}_{i,t,t+h}], \tilde{hl}_{i,t}] + Cov[\mathbb{F}_t[\tilde{pe}_{i,t,t+h}], \tilde{hl}_{i,t}] \quad (25)$$

where $Var[\cdot]$ and $Cov[\cdot]$ are cross-sectional variances and covariances. Dividing both sides by $Var[\tilde{hl}_{i,t}]$ yields the variance decomposition:

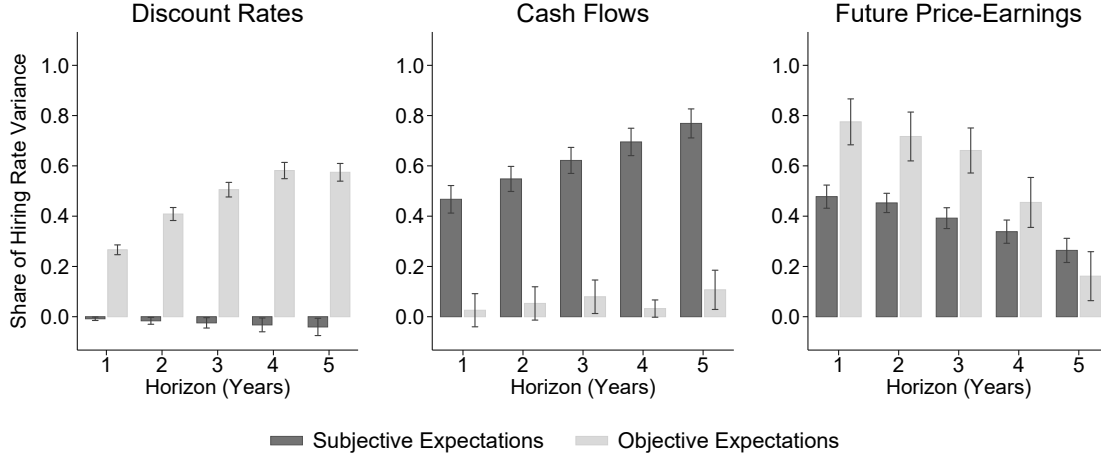
$$1 = - \underbrace{\frac{Cov[\mathbb{F}_t[\tilde{r}_{i,t,t+h}], \tilde{hl}_{i,t}]}{Var[\tilde{hl}_{i,t}]}}_{\text{Discount Rate News}} + \underbrace{\frac{Cov[\mathbb{F}_t[\tilde{e}_{i,t,t+h}], \tilde{hl}_{i,t}]}{Var[\tilde{hl}_{i,t}]}}_{\text{Cash Flow News}} + \underbrace{\frac{Cov[\mathbb{F}_t[\tilde{pe}_{i,t,t+h}], \tilde{hl}_{i,t}]}{Var[\tilde{hl}_{i,t}]}}_{\text{Future Price-Earnings News}} \quad (26)$$

The sample covers all common stocks (share code 10 and 11) listed on NYSE, AMEX, and NASDAQ, restricted to firms that have sufficient data to construct total employee counts (EMP) from Compustat and the median analyst stock return and earnings growth forecasts at the five-year horizon from IBES, as described in Section 2.

Figure 6 shows that under subjective expectations, cross-sectional dispersion in the hiring rate is dominated by differences in expected cash flows. At the five-year horizon, 79.2% of the cross-sectional variance is explained by the expected cash flows. In contrast, only -5.2% of the variation is explained by discount rates, and 29.8% is attributed to differences in the terminal future price-earnings expectation. Under subjective beliefs, a higher discount rate is associated with a higher hiring rate, a direction that is not consistent with the predictions of the search model. These results indicate that firms across the distribution of idiosyncratic shocks have sharply different expectations about future cash flows when expectations are subjective, and these differences in beliefs translate into differences in perceived hiring incentives. The combined contribution of the three components sums to 103.8% at the five-year horizon, a value close to 100.0% suggesting that the approximations used in the decomposition are reasonably accurate despite being freely estimated without imposing this constraint.

Under objective expectations, 60.2% of cross-sectional variation in hiring is explained by differences in expected discount rates at horizon five, while only 13.1% is explained by expected

Figure 6: Cross-Sectional Decomposition of the Hiring Rate



Notes: Figure illustrates the discount rate, cash flow, and future price-earnings components from the cross-sectional decomposition of the hiring rate. Light bars show contributions under objective expectations; dark bars show contributions under subjective expectations. Subjective expectations $\mathbb{F}_t$ are constructed from IBES analyst forecasts (discount rates and cash flows). Objective expectations $\mathbb{E}_t$ are based on machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. x -axis denotes the forecast horizon h . The sample is quarterly from 2005Q1 to 2023Q4. Each bar shows 95% confidence intervals under two-way clustering by firm and time.

cash flows. This pattern is consistent with existing estimates showing that, under objective expectations, much of the variation across firms in asset valuations comes from dispersion in risk premia rather than expected cash flows (De La O et al., 2024).

Under subjective beliefs, cross-sectional variation in hiring is driven primarily by firms overreacting to news about future cash flows. This provides a micro-foundation for the aggregate results: the same belief distortion that drives aggregate unemployment operates at the firm level, where hiring decisions are made.²⁴

Employment Response to Idiosyncratic Shocks Objective and subjective beliefs imply qualitatively different responses to idiosyncratic shocks. Under rational expectations with adjustment costs, positive shocks leave profits per worker stable or temporarily higher; under systematic overreaction, firms overhire relative to the benchmark, depressing profits per worker. Figure 7 tests these competing predictions with local projections under two specifications. For the impulse response under subjective beliefs, I estimate:

$$y_{i,t+h} = \alpha_i^{\mathbb{F}} + \tau_{s(i),t}^{\mathbb{F}} + \beta^{\mathbb{F}}(\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]) + \gamma^{\mathbb{F}'} \mathbf{X}_{i,t} + \epsilon_{i,t+h}^{\mathbb{F}} \quad (27)$$

²⁴The cross-sectional results suggest that much of the dispersion in hiring rates reflects belief-driven forecast errors. Such distorted expectations can act as a wedge that misallocates labor by inducing overhiring at optimistic firms and underhiring at pessimistic firms (Ma et al., 2020; David et al., 2022; Ropele et al., 2024).

where $y_{i,t}$ denotes the outcome variable (log profits per worker $\log(E_{i,t+h}/L_{i,t+h})$, employment $\log(L_{i,t+h})$, or profits $\log(E_{i,t+h})$), and $\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]$ captures subjective forecast revisions for earnings growth. For the impulse response implied by objective beliefs, I estimate a similar local projection while replacing the survey forecast revision with the corresponding machine forecast revision. To directly test the role of belief distortions, I estimate:

$$y_{i,t+h} = \alpha_i^{\mathbb{D}} + \tau_{s(i),t}^{\mathbb{D}} + \beta^{\mathbb{D}}(\mathbb{D}_{i,t}[\Delta e_{i,t+1}] - \mathbb{D}_{i,t-1}[\Delta e_{i,t+1}]) + \gamma^{\mathbb{D}'}\mathbf{X}_{i,t} + \varepsilon_{i,t+h}^{\mathbb{D}} \quad (28)$$

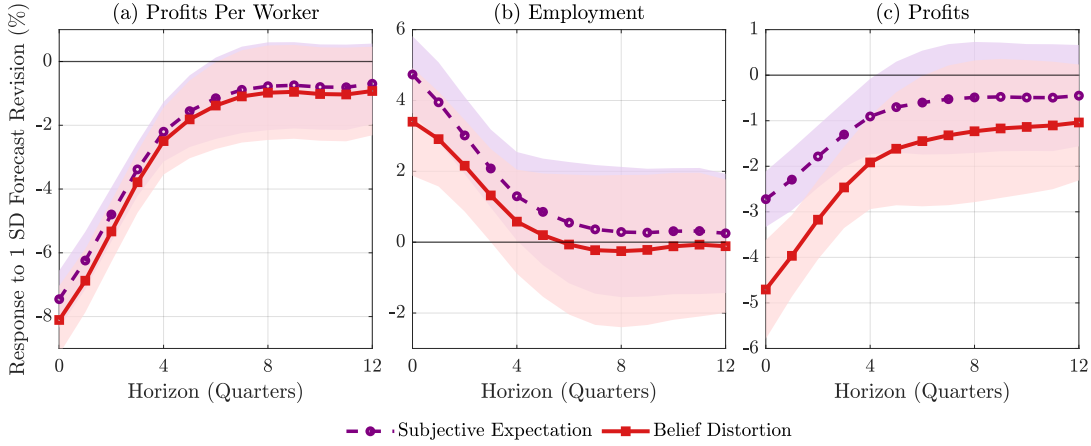
where $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] \equiv \mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{E}_t[\Delta e_{i,t+1}]$ denotes the belief distortion in earnings growth forecasts, defined as the wedge between subjective expectations and objective machine learning expectations. The regressor $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] - \mathbb{D}_{i,t-1}[\Delta e_{i,t+1}]$ captures the revision in belief distortion, measuring changes in how much firms over- or under-estimate earnings growth relative to the objective benchmark. As documented in Figure 1, these belief distortions exhibit overreaction to idiosyncratic shocks. Both specifications include firm fixed effects α_i , industry-time fixed effects $\tau_{s(i),t}$ where $s(i)$ is firm i 's 2-digit SIC industry, allowing firms in different industries to have different exposures to aggregate risk. Controls $\mathbf{X}_{i,t}$ contain lags of the dependent variable and forecast revisions. The comparison of $\beta^{\mathbb{F}}$ versus $\beta^{\mathbb{D}}$ provides a direct test of belief distortions: if systematic overreaction drives the response, then $\beta^{\mathbb{F}}$ and $\beta^{\mathbb{D}}$ should be similar in magnitude.

Figure 7 reports the impulse responses estimated from (27) and (28). Across all three outcomes (profits per worker in panel a, employment in panel b, and total profits in panel c), the responses to the survey revision (violet) and to the belief-distortion revision (red) are nearly identical: profits per worker decline, employment rises sharply, and total profits fall slightly. The close alignment between the two responses indicates that the hiring response to idiosyncratic shocks is driven almost entirely by the distortion component of subjective beliefs.²⁵ The belief distortion revision alone explains 50% of the variation in contemporaneous profits per worker ($R^2 = 0.50$), accounting for 82% of the total variation explained by subjective forecast revisions ($R^2 = 0.61$). This quantitative decomposition confirms that the distortionary component of subjective beliefs drives the observed dynamics in profits per worker.

Following a positive idiosyncratic shock, firms with distorted beliefs become overly optimistic, post more vacancies, and expand employment aggressively. The rapid increase in the wage bill and hiring costs (vacancy posting, recruiting, training) outpaces any revenue gains from the shock, leading to a decline in total profits. Because idiosyncratic shocks are transitory, the

²⁵The corresponding impulse response under objective expectations (machine learning forecasts) shows profits per worker remaining stable following idiosyncratic shocks, as shown in Appendix Figure A.3. This stability is consistent with rational expectations models where firms correctly identify idiosyncratic shocks as transitory and diversifiable, and therefore do not overreact to the shock.

Figure 7: Employment Response to Idiosyncratic Forecast Revision



Notes: Figure displays impulse responses to earnings growth forecast revisions, estimated via local projections with firm and time fixed effects. Panel (a): log profits per worker $\log(E_{i,t+h}/L_{i,t+h})$. Panel (b): log employment $\log(L_{i,t+h})$. Panel (c): log profits $\log(E_{i,t+h})$. Violet line: impulse response under subjective expectations, where survey forecast revision is the earnings growth revision $\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]$. Red line: impulse response to belief distortion revision $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] - \mathbb{D}_{i,t-1}[\Delta e_{i,t+1}]$, where $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] \equiv \mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{E}_t[\Delta e_{i,t+1}]$. Observations weighted by each firm's market value. Shaded areas: two-way clustered 95% confidence intervals by firm and time. Data sample: 1999Q1–2023Q4.

productivity improvement fades quickly while employment remains persistently elevated due to labor market frictions. By the time new workers are hired and on the payroll, the productivity benefit has dissipated, leaving firms with elevated labor costs but no offsetting revenue increase. The result is both declining total profits and sharply declining profits per worker, demonstrating that belief distortions drive fluctuations in hiring decisions.²⁶ The magnitude of these responses is economically meaningful. The peak increase in employment represents roughly ten percent of the cross-sectional dispersion in firm-level employment growth, placing the response well within the range of differences observed across firms.

Under objective expectations, however, firms recognize that idiosyncratic earnings shocks are typically short-lived, with limited persistence. Given labor market frictions, including vacancy posting costs, recruiting and training expenses, and separation costs, hiring represents a costly investment whose payoff materializes only gradually. A firm with objective beliefs would therefore anticipate that by the time new workers are hired and fully productive, a transitory shock may have already dissipated, leaving little benefit from expanding employment. Moreover, the option value of waiting to observe whether a shock persists further dampens the incentive to respond immediately. As a result, while some hiring response to idiosyncratic shocks remains optimal,

²⁶The profit measure is IBES street earnings, which corresponds to adjusted net income excluding one-off charges but including depreciation. Since capital expenditures are capitalized rather than expensed, they affect earnings gradually through depreciation. In the short run, investment does not directly depress IBES earnings, so the observed decline in profits per worker reflects a strong response in hiring rather than capital investment patterns.

objective firms should exhibit substantially muted responses relative to permanent shocks, with the optimal response declining as adjustment costs increase. In contrast, the empirical evidence shows that firms with distorted beliefs exhibit large hiring responses to idiosyncratic shocks that subsequently reduce profits per worker, consistent with belief-driven overreaction rather than rational adjustment.

Stock Return Response to Idiosyncratic Shocks If the same distortions that lead firms to adjust their hiring excessively also get priced into stock valuations, then subjective cash flow revisions should predict returns with an initial overshoot followed by disappointment, while objective discount rate revisions should not predict returns since idiosyncratic shocks are diversifiable. To test this prediction, I estimate local projections of annual stock returns on forecast revisions. For the impulse response under subjective beliefs, I estimate:

$$r_{i,t+h} = \alpha_i^{\mathbb{F}} + \tau_{s(i),t}^{\mathbb{F}} + \beta^{\mathbb{F}}(\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]) + \gamma^{\mathbb{F}'}\mathbf{X}_{i,t} + \varepsilon_{i,t+h}^{\mathbb{F}} \quad (29)$$

where $r_{i,t+h}$ denotes the one-year stock return from t to $t+h$. I use the revision in the machine forecast of stock returns to estimate similar impulse responses under objective beliefs. To isolate the role of belief distortions, I estimate:

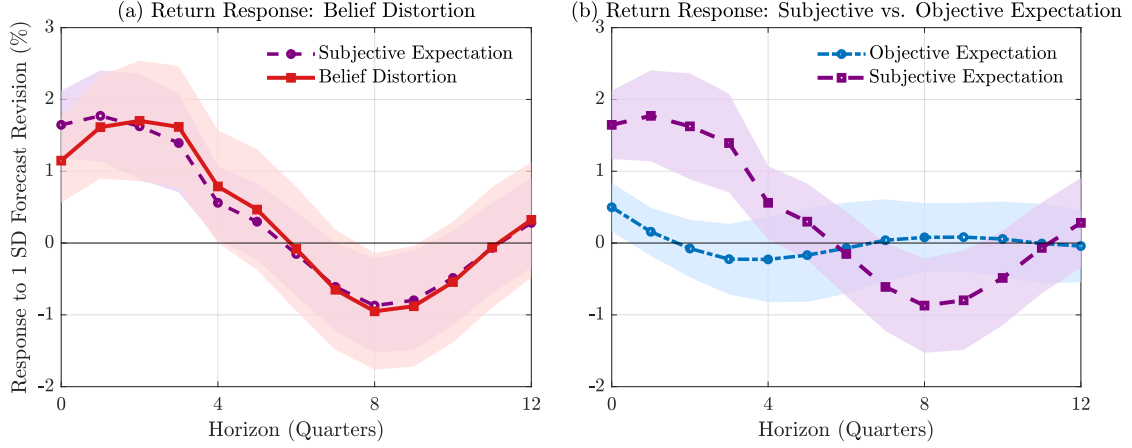
$$r_{i,t+h} = \alpha_i^{\mathbb{D}} + \tau_{s(i),t}^{\mathbb{D}} + \beta^{\mathbb{D}}(\mathbb{D}_{i,t}[\Delta e_{i,t+1}] - \mathbb{D}_{i,t-1}[\Delta e_{i,t+1}]) + \gamma^{\mathbb{D}'}\mathbf{X}_{i,t} + \varepsilon_{i,t+h}^{\mathbb{D}} \quad (30)$$

where $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] \equiv \mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{E}_t[\Delta e_{i,t+1}]$ denotes the belief distortion. Figure 8 reports the resulting impulse responses. Under subjective beliefs, positive forecast revisions generate a short-run increase in realized returns as optimistic investors bid up prices, followed by a reversal as the initial overreaction is corrected. In contrast, under objective (machine) expectations, the response of returns to idiosyncratic forecast revisions is statistically insignificant at all horizons, consistent with rational models in which firm-specific shocks are transitory and diversifiable. The implied return response is large in economic terms and accounts for close to ten percent of the cross-sectional dispersion in stock returns, highlighting that belief distortions amplify movements in both employment and asset prices. The belief-distortion revision alone explains about 62% of the variation in contemporaneous stock returns, mirroring its role for profits per worker.

7 A General-Equilibrium Model of Belief Distortions, Asset Prices, and Unemployment

The empirical sections established two facts: firms' cash-flow forecasts overreact to recent news, and this overreaction moves hiring. This section builds a general-equilibrium search-and-matching

Figure 8: Stock Return Response to Idiosyncratic Forecast Revision



Notes: Figure displays impulse responses to earnings growth forecast revisions, estimated via local projections with firm and time fixed effects. Panel (a): Impulse response comparing subjective beliefs and belief distortions. Panel (b): Impulse response comparing subjective and objective beliefs. Blue line: impulse response under objective expectations, where machine forecast revision is the earnings growth revision $\mathbb{E}_t[\Delta e_{i,t+1}] - \mathbb{E}_{t-1}[\Delta e_{i,t+1}]$. Violet line: impulse response under subjective expectations, where survey forecast revision is the earnings growth revision $\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]$. Red line: impulse response to belief distortion revision $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] - \mathbb{D}_{i,t-1}[\Delta e_{i,t+1}]$, where $\mathbb{D}_{i,t}[\Delta e_{i,t+1}] \equiv \mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{E}_t[\Delta e_{i,t+1}]$. Observations weighted by each firm's market value. Shaded areas: two-way clustered 95% confidence intervals by firm and time. Data sample: 1999Q1–2023Q4.

model that reproduces both facts and follows them through to asset prices and unemployment. A single representative family owns the firms and prices their equity with Epstein-Zin preferences. Because only part of its consumption is exposed to firm dividends, which is what limited participation means here, belief-driven swings in cash flows show up in the pricing kernel.

Two belief distortions drive the model, and I measure each one directly. The first is the overreaction itself: firms overreact to recent news about their cash-flow growth but hold correct views about how persistent that growth is. Priced by the family's discount factor, this belief generates the equity premium and the swings in valuations, and it raises and lowers the value a firm places on a new hire. Keeping the overreaction separate from the perceived persistence is what lets one belief match two survey facts at once, the strong short-run overreaction and the high subjective cash-flow share, which a belief that simply over-extrapolates persistence cannot. The second distortion is a rigid wage: the wage workers expect to earn if they change jobs does not move with the cycle, so when the value of hiring rises the wage does not keep up, the profit from a filled job widens, and the firm posts more vacancies. Turning both beliefs off returns the model to rational expectations, where the wage is flexible and unemployment barely moves, as in Shimer (2005). Comparing the two economies channel by channel is how I measure what the distortions contribute: the cash-flow belief drives the volatility of asset prices, and the wage belief drives the volatility of unemployment. The two act on different objects and pull in opposite directions over the cycle, yet they come from the same habit of mind, overreacting to

salient aggregate news while anchoring one's own wage to its recent level; Appendix A.4 makes this precise and shows why one parameter cannot do both jobs.

The model is solved at the monthly frequency and all moments are aggregated to the annual frequency at which they are measured. Section 7.1 states the environment. Sections 7.2-7.7 state the optimization problem of each agent and the belief structure. Section 7.8 gives the aggregation and Section 7.9 the equilibrium definition. Section 7.11 derives the stochastic discount factor and the value of hiring. Section 7.12 states and proves the decomposition results. Section 7.13 calibrates every parameter and Section 7.14 reports the quantitative results.

7.1 Environment

Time is discrete and the period is one month. There is a unit continuum of firms indexed by $i \in [0, 1]$, each of which is a single job that can be filled or vacant. Firms are owned by a single representative family whose members pool consumption and who supply labor. A fraction L_t of the family's members are employed and the remaining $U_t = 1 - L_t$ are unemployed and searching.

Productivity. Log aggregate labor productivity a_t grows at a rate that contains a small persistent component x_t , the long-run-risk state of Bansal and Yaron (2004):

$$\Delta a_{t+1} = \mu + x_t + \sigma_t \varepsilon_{t+1}, \quad (31)$$

$$x_{t+1} = \rho x_t + \varphi_e \sigma_t \eta_{t+1}, \quad (32)$$

$$\sigma_{t+1}^2 = \bar{\sigma}^2 + \nu_\sigma (\sigma_t^2 - \bar{\sigma}^2) + \sigma_w w_{t+1}, \quad (33)$$

with ε, η, w standard normal and mutually independent under the objective measure. The state x_t is the expected growth of productivity; σ_t^2 is stochastic variance. Two objective sources of discount-rate variation coexist: a time-varying risk premium through σ_t^2 , and belief-driven return predictability through the over-extrapolation of x_t (Proposition 2). Their separate contributions are reported in Section 7.14.

Firm-level productivity. Firm i operates one job with log productivity $a_{i,t}$ that shares the aggregate trend and carries an idiosyncratic expected-growth component. Its growth is

$$\Delta a_{i,t+1} = \mu + x_{i,t} + \sigma_t \varepsilon_{i,t+1}, \quad x_{i,t} = x_t + \xi_{i,t}, \quad \xi_{i,t+1} = \rho_\xi \xi_{i,t} + \sigma_\xi e_{i,t+1}, \quad (34)$$

where $\xi_{i,t}$ is a mean-zero idiosyncratic expected-growth state, independent across firms and of the aggregate shocks, with persistence $\rho_\xi < \rho$ (firm-specific growth reverts faster than aggregate growth) and ε_i, e_i standard normal. Splitting firm expected growth into the common x_t and the

idiosyncratic $\xi_{i,t}$ is what makes the aggregate the integral of the firm block: integrating the firm value of hiring over i averages out $\xi_{i,t}$ and leaves the common x_t (Section 7.8).

Cash flows and operating leverage. A filled job at firm i produces flow output $Y_{i,t} = e^{a_{i,t}}$, pays a wage $W_{i,t}$, and incurs a quasi-fixed operating cost $F e^{\mu t}$ that does not scale with the match's contemporaneous productivity (overhead, capital services, and non-labor inputs). The flow profit, or dividend, is the residual

$$D_{i,t} = e^{a_{i,t}} - W_{i,t} - F e^{\mu t}, \quad (35)$$

which is the firm's cash flow, paid out to the family. The fixed cost is what makes the dividend a small, volatile residual: it is the source of operating leverage in the sense of Novy-Marx (2011), and it is why the empirically relevant profit share is the firm's *operating margin* (profit over sales), which nets out non-labor costs, rather than one minus the labor share. With a steady-state profit share $\bar{s}h = D/Y$ equal to the operating margin, the log deviation of $D_{i,t}$ is a levered function of output and the wage (Lemma 1). Because $a_{i,t}$ inherits the diagnostic expected growth $\tilde{x}_{i,t}$, so does its dividend, and the value of a claim on that dividend rises and falls with the belief, amplified because the claim is long-lived (Section 7.11).

7.2 Households

The family. A single representative family pools the consumption of its members, as in Merz (1995) and Andolfatto (1996). Of a unit mass of members, L_t are employed at wage W_t and $U_t = 1 - L_t$ are unemployed and each produce home output $b e^{a_t}$. The family owns the entire measure of firms and receives their flow profits net of vacancy costs. Firm equity is the only store of value, so in equilibrium the family holds the market and consumption equals net output (Section 7.9).

Risk sharing, timing, and balanced growth. The family fully shares employment risk but not equity risk, which is the source of the limited participation below. The timing convention and the balanced-growth normalization are stated in Appendix A.11.

Preferences. The family ranks consumption streams with the recursive Epstein-Zin aggregator of Epstein and Zin (1989), evaluated under its *subjective* measure $\mathbb{F}$ (Section 7.7):

$$V_t = \left[(1 - \beta) C_t^{1-1/\psi} + \beta \left(\mathbb{E}_t^{\mathbb{F}} [V_{t+1}^{1-\gamma}] \right)^{\frac{1-1/\psi}{1-\gamma}} \right]^{\frac{1}{1-1/\psi}}, \quad (36)$$

with relative risk aversion γ , elasticity of intertemporal substitution ψ , discount factor β , and $\theta_{EZ} \equiv (1 - \gamma)/(1 - 1/\psi)$.

The family's problem. The family chooses consumption C_t and firm-portfolio holdings S_{t+1} (priced at Q_t , paying aggregate dividend $D_t = \int_0^1 D_{i,t} L_{i,t} di$) to maximize (36) subject to the budget constraint $C_t + Q_t S_{t+1} = W_t L_t + b e^{a_t} U_t + (Q_t + D_t) S_t$, taking the wage, the employment law (43), the price Q_t , and its subjective measure as given. The Euler equation $1 = \mathbb{E}_t^{\mathbb{F}}[M_{t+1}(Q_{t+1} + D_{t+1})/Q_t]$ holds with the Epstein-Zin stochastic discount factor

$$M_{t+1} = \beta^{\theta_{EZ}} \left(\frac{C_{t+1}^p}{C_t^p} \right)^{-\theta_{EZ}/\psi} (R_{t+1}^W)^{\theta_{EZ}-1}, \quad (37)$$

where R_{t+1}^W is the return on the family's wealth portfolio and C^p is the consumption of the marginal member defined next. Equation (37) prices every claim in the economy, including the value of a filled job (40).

Limited participation. The consumption that enters the kernel (37) is that of the marginal member who holds the firm equity, not necessarily average family consumption. Members differ in the composition of their income: a share of marginal consumption is financed out of the volatile dividend stream and the rest out of smooth labor income. Writing $s_D \in [0, 1]$ for the dividend share of the marginal member's consumption,

$$\Delta \log C_t^p = (1 - s_D) \Delta \log C_t^\ell + s_D \Delta \log D_t, \quad (38)$$

with C^ℓ the labor-income-financed component and aggregate consumption C_t satisfying goods-market clearing (Section 7.9). Section 7.3 shows how s_D arises from a participation friction and what it is disciplined by; here it is the reduced-form share that maps firm cash flows into the pricing kernel. Log-linearizing (37) gives a log kernel whose prices of risk, $\lambda_e = \gamma$ on the consumption shock and λ_η, λ_w on the long-run-risk and volatility shocks, are functions of (γ, ψ) and the wealth-consumption loadings rather than free parameters (Appendix A.18). Because C_t^p loads on the growth belief through s_D and the dividend, the same x_t the family over-extrapolates enters the kernel and is priced.

7.3 The limited-participation share

The share s_D in (38) is the equity exposure of the marginal investor, not a free wedge on the kernel, and it is disciplined by one asset-pricing moment. Consider a participation structure of the form in Mankiw and Zeldes (1991) and Guvenen (2009): a fraction of the family's members

pay a fixed cost to hold the firm equity and price it, while the rest hold only a claim to labor income. The marginal member who prices equity consumes his labor income plus the dividends on his concentrated equity position, so his consumption growth is (38) with s_D equal to the dividend share of that consumption. Two implications follow. First, s_D is a primitive of the participation structure, the equity exposure of the marginal investor, not a preference or belief parameter; and empirically the marginal stockholder's consumption is far more exposed to equity than the representative household's, so an s_D well above the aggregate dividend-to-consumption ratio is the relevant case. Second, with the prices of risk (A.11) fixed by (γ, ψ) and the growth belief, s_D maps monotonically to the level of the equity premium: it is the single parameter that translates the volatility of firm cash flows into the volatility of the pricing kernel. The share s_D can be disciplined two ways, and they agree. Set to the equity premium, it is 0.12; disciplined instead by external stockholder-consumption data, it is close to the same value. With aggregate consumption growth volatility of about 0.4 percent per year and dividend growth volatility of about 5.5 percent (the operating-leverage residual of Lemma 1), $s_D = 0.12$ makes the marginal investor's consumption growth about twice as volatile as aggregate consumption, which is inside the range Malloy et al. (2009) estimate for stockholder consumption relative to per-capita consumption. So s_D is not identified by the premium alone: it is pinned by the exposure of stockholder consumption to the market, an object measured outside the model, and it then delivers a five-percent premium. At that fixed s_D the belief raises the premium from the rational 0.9 percent to 5 percent, so the belief does the covariance work and s_D scales it. With $s_D = 0$ the representative-family kernel prices only smooth aggregate consumption and delivers a premium of 2.8 percent; limited participation is load-bearing, and it is disciplined rather than free.

Microfoundation. The participation cutoff that yields $s_D = \omega_D/\Pi$, and the consistency between the marginal participant's and aggregate consumption, are in Appendix A.6.

7.4 Firms and the value of hiring

A filled job of firm i generates flow profit equal to output minus the wage. With a constant-returns match technology and a small profit share, the log deviation of the flow profit per unit of productivity responds to the productivity-wage gap $g_{i,t} = \log(A_{i,t}/W_{i,t})$, where $A_{i,t} \equiv e^{a_{i,t}}$ is the level of firm productivity, with operating leverage.

Lemma 1 (Operating leverage). *Let the steady-state profit share be the operating margin $\overline{sh} =$*

D/Y , with the wage bill a share $sh_W = W/Y$ and the fixed cost a share $sh_F = F/Y = 1 - \bar{sh} - sh_W$ of output. Then to first order the log deviation of the dividend is

$$\widehat{D}_{i,t} = \frac{1}{\bar{sh}} \widehat{Y}_{i,t} - \frac{sh_W}{\bar{sh}} \widehat{W}_{i,t} = \frac{1}{\bar{sh}} (\widehat{Y}_{i,t} - sh_W \widehat{W}_{i,t}), \quad (39)$$

so the dividend responds to output with elasticity $1/\bar{sh}$, the degree of operating leverage. The fixed cost drops out of the deviation but sets the small level $\bar{sh}$ that magnifies it.

The measured operating margin $\bar{sh} = 0.088$ (Section 7.13) gives an operating leverage $1/\bar{sh} = 11.4$: a one-percent movement in output moves the dividend by about eleven percent. This is the object the operating-margin data discipline, and it is why the profit share that governs earnings volatility is the small operating margin, not the larger complement of the labor share; the difference is the fixed and non-labor costs, which the operating margin nets out and which the dividend claim in (35) contains. The proof is in Appendix A.2.

The Bellman equation of a filled job. Let $J_{i,t}$ be the value to the family of a filled job at firm i , measured in units of the family's marginal utility and priced by the kernel (37). It solves

$$J_{i,t} = D_{i,t} + (1 - \delta_t) \mathbb{E}_t^{\mathbb{F}}[M_{t+1} J_{i,t+1}], \quad (40)$$

where δ_t is the (possibly countercyclical) separation rate. The value of a filled job is a long-duration claim on the firm's cash flows, discounted by the family's kernel under its subjective measure.

Vacancy creation and free entry. A firm may post a vacancy at flow cost κe^{a_t} , which meets a worker with probability $q(\theta_t)$ per period, where $\theta_t = v_t/U_t$ is market tightness. Free entry drives the value of a vacancy to zero, giving the free-entry condition

$$\kappa e^{a_t} = q(\theta_t) \underbrace{\mathbb{E}_t^{\mathbb{F}}[M_{t+1} J_{i,t+1}]}_{\equiv P_{i,t}}. \quad (41)$$

Equations (40)-(41) are the firm's optimization: post a vacancy whenever the expected discounted value of the filled job $P_{i,t}$ covers the hiring cost. Since $q(\theta_t) = \chi \theta_t^{-\alpha}$, (41) is the job-creation condition

$$\theta_t^\alpha = \frac{\chi}{\kappa} e^{-a_t} P_{i,t}, \quad (42)$$

which ties tightness, and hence the filling rate and unemployment, to the value of hiring $P_{i,t}$. In log deviations $\alpha \widehat{\theta}_t = \widehat{P}_{i,t}$: a one-percent rise in the value of hiring raises tightness by $1/\alpha$

percent, so the volatility of unemployment inherits the volatility of the value of hiring. Iterating (40) forward, the value of hiring is a long-duration claim whose loading on the growth belief is the growth-stock duration Λ of Section 7.11 (Appendix A.7).

7.5 The labor market

New matches are formed by a constant-returns matching function $m_t = \chi U_t^\alpha v_t^{1-\alpha}$, which gives a job-finding rate $f(\theta_t) = m_t/U_t = \chi \theta_t^{1-\alpha}$ and a vacancy-filling rate $q(\theta_t) = m_t/v_t = \chi \theta_t^{-\alpha}$, with $f = \theta q$. Employment evolves as

$$L_{t+1} = (1 - \delta_t) L_t + f(\theta_t) (1 - L_t). \quad (43)$$

Steady state and the Beveridge curve. In the deterministic steady state the flow into unemployment equals the flow out, $\bar{\delta} \bar{L} = \bar{f} (1 - \bar{L})$, so the unemployment rate is $\bar{U} = 1 - \bar{L} = \bar{\delta}/(\bar{\delta} + \bar{f})$; at the calibrated monthly $\bar{\delta} = 0.028$ and $\bar{f} = 0.45$ this is $\bar{U} = 0.058$. The Beveridge curve is the locus of $(\bar{U}, \bar{v})$ consistent with the matching function, $\bar{v} = \bar{U} \bar{\theta}$ with $\bar{\theta} = (\bar{f}/\chi)^{1/(1-\alpha)}$; its negative slope, unemployment and vacancies moving in opposite directions over the cycle, is the correlation reported in Section 7.14. The elasticity of tightness to the value of hiring implied by (42) is $1/\alpha$, which governs how far a movement in $\hat{P}_{i,t}$ travels into $\hat{\theta}_t$.

The separation margin. In the baseline the separation rate is a constant, $\delta_t = \bar{\delta}$. In the two-margin version it also has a countercyclical component that rises when the value of hiring falls, $\delta_t = \bar{\delta}(1 - \phi_\delta \hat{P}_t)$, with the elasticity ϕ_δ disciplined by the one-quarter separation share of Shimer (2012): about one quarter of the cyclical variation in unemployment operates through separations rather than hiring. This adds a second channel through which the belief-driven value of hiring moves unemployment and is the source of the additional amplification reported in Appendix A.5.

Frequency. Because a single job spell turns over within the year at the monthly job-finding rate $\bar{f} = 0.45$, the matching block is solved monthly and aggregated to the year; solved directly at the annual frequency the flow equation (43) is degenerate, with a steady-state unemployment rate near one third and an unstable annual law of motion (Appendix A.5). All labor moments are computed on the annual series obtained by time-aggregating the monthly simulation.

7.6 Wage determination

The wage splits the match surplus by Nash bargaining. Let $\mathcal{W}_{i,t}$ be the value to a worker of being employed at firm i and $\mathcal{U}_t$ the value of unemployment, both to the family and priced by its kernel:

$$\mathcal{W}_{i,t} = W_{i,t} + \mathbb{E}_t^{\mathbb{F}}[M_{t+1}((1 - \delta_t)\mathcal{W}_{i,t+1} + \delta_t \mathcal{U}_{t+1})], \quad (44)$$

$$\mathcal{U}_t = b e^{a_t} + \mathbb{E}_t^{\mathbb{F}}[M_{t+1}(f(\theta_t) \mathcal{W}_{t+1}^{\text{new}} + (1 - f(\theta_t)) \mathcal{U}_{t+1})], \quad (45)$$

where $\mathcal{W}_{t+1}^{\text{new}}$ is the value of a newly formed job. The worker's surplus is $\mathcal{S}_{i,t}^W = \mathcal{W}_{i,t} - \mathcal{U}_t$ and the firm's surplus is the value of the filled job $J_{i,t}$, since the value of a vacancy is zero by free entry (41). The wage solves the Nash program

$$W_{i,t} = \arg \max_W (\mathcal{S}_{i,t}^W)^{\eta_w} (J_{i,t})^{1-\eta_w}, \quad (46)$$

whose first-order condition is the surplus-sharing rule $\eta_w J_{i,t} = (1 - \eta_w) \mathcal{S}_{i,t}^W$: the worker receives a fraction η_w of the total match surplus. Substituting the value functions and the free-entry condition and rearranging gives the wage

$$W_{i,t} = \eta_w (e^{a_{i,t}} + \kappa e^{a_t} \theta_t) + (1 - \eta_w) z_t^w, \quad (47)$$

where $e^{a_{i,t}}$ is the marginal product, $\kappa e^{a_t} \theta_t$ is the firm's foregone search cost that the tight-market worker appropriates, and z_t^w is the worker's outside option, equal to home production plus the continuation term in (45). The forward-looking component of z_t^w is the present value the worker attaches to the surplus of the next job, and up to the discount factor its cyclical part is the wage the worker expects to earn upon re-employment, $\mathbb{F}_t^w[W_{t+1}^{\text{new}}]$. This is the survey object measured below: expected re-employment wages, not realized wages, are what enter the threat point.

The worker's belief. The wage rigidity is itself a measured belief distortion, not a mechanical friction: it is the equilibrium consequence of the worker's forecast of the re-employment wage, read from surveys. The worker forecasts the re-employment wage under a subjective measure $\mathbb{F}^w$ that anchors the wage to a slow-moving status-quo norm, a reference-dependent belief in the sense of Kőszegi and Rabin (2006) and the anchoring half of the cognitive family of Section 7.7. Writing the forecast as a weight ϕ on the norm and $1 - \phi$ on the rational forecast gives $\beta^w = 1 - \phi$, so the worker does not extrapolate the current cycle into $\mathbb{F}_t^w[W_{t+1}^{\text{new}}]$, which moves little with current productivity and $\beta^w \approx 0$. Under rational expectations the worker would track the realized cyclical wage, $\beta^w = 1$; the wedge is what the surveys measure directly, by comparing expected wage growth (acyclical) to realized wage growth (countercyclical).

The model thus contains two measured belief distortions, and it is worth being precise that they are not the same object and cannot be collapsed into one. They are *opposite-signed on the cycle*. The growth belief *over*-reacts to the current innovation ($\zeta > 0$ in (49), the source of the Coibion-Gorodnichenko overreaction and the asset-price volatility); the wage belief *under*-reacts, the expected outside option failing to track the cycle ($\beta^w \approx 0$). A single diagnostic parameter cannot produce both, because a worker who over-perceives aggregate growth would over-perceive the value of the next job as well, raising the threat point in booms and making the wage *more* flexible. Feeding the firm's growth belief into the worker's threat point makes this quantitative: the outside option's cyclicalitity would be $\beta^w = 1.39$ (it over-responds), the wage elasticity $\omega = 1.35$ (hyper-flexible), and the unemployment volatility would collapse to $SD(U) = 0.14$, near the rational-expectations benchmark, rather than reproducing the data. What the two beliefs share is narrower than a common primitive: both are departures from rational expectations measured from surveys rather than assumed, and both operate through the same object, the value of a filled job. But one over-extrapolates the aggregate and the other anchors the worker's own prospect, and it is the combination, not either alone, that matches the joint evidence.

The allocative versus the realized new-hire wage. The rigidity is in the allocative wage, the worker's expected outside option in the threat point, not in the realized new-hire wage, which is flexible through the procyclical tightness term; this reconciles the model with Pissarides (2009) and Haefke et al. (2013) (Appendix A.16).

Lemma 2 (Wage cyclicalitity). *Log-linearizing (47) around the deterministic steady state, the elasticity of the wage to productivity is*

$$\omega \equiv \frac{d \log W_t}{d \log A_t} = \eta_w + (1 - \eta_w) \beta^w, \quad (48)$$

where β^w is the cyclicalitity of the worker's expected outside option.

Under rational expectations the expected outside option tracks realized market wages, $\beta^w \approx 1$, and the wage is flexible, $\omega \approx 1$. In the data the expected wage is acyclical: regressing expected wage growth on the change in unemployment gives slopes of +0.003 (Survey of Consumer Expectations), +0.001 (CFO survey), and +0.005 (Livingston), against realized slopes of -0.048 and -0.062 , a ratio of 0.05 to 0.10. Hence $\beta^w \approx 0$ and $\omega = \eta_w \in [0.05, 0.10]$: the wage is rigid because the worker's *expected* outside option does not move with the cycle, an object measured from surveys rather than assumed. Both the wage cyclicalitity ω and the operating leverage that governs

unemployment volatility are measured externally (Section 7.13), the latter from the Compustat operating margin, so the level of unemployment volatility is a prediction, not a target.

7.7 Beliefs

The model contains two belief distortions, the family’s belief about aggregate cash-flow growth developed here and the worker’s belief about the own re-employment wage of Section 7.6. They are two manifestations of a single behavioral primitive: agents *over-weight salient aggregate innovations* and *anchor their own status-quo position*. Applied to aggregate growth, over-weighting the latest innovation is a diagnostic belief in the sense of Bordalo et al. (2018); applied to the worker’s own wage, anchoring to a slow-moving norm is reference-dependence in the sense of Kőszegi and Rabin (2006). The same disposition thus produces opposite responses to the cycle depending on the object: the aggregate is *over*-extrapolated (the source of the overreaction and the asset-price volatility), while the worker’s own wage is *under*-extrapolated (the source of the rigidity). This is not a coincidence of two unrelated frictions but one cognitive family read off two independent surveys; the formal statement, and the demonstration that a single parameter cannot substitute for the two because the required signs differ, are in Appendix A.4. The remainder of this section develops the aggregate side.

I now formalize the family’s belief about cash-flow growth. It overreacts to the most recent news about growth but keeps a correct view of how persistent that growth is. That split is what lets one belief match both survey facts at once, the strong overreaction the paper documents and the high subjective cash-flow share; a belief that instead over-extrapolates persistence cannot match both.

The diagnostic belief. The objective expected-growth state is $x_t = \rho x_{t-1} + \varphi_e \sigma_{t-1} \eta_t$ with innovation η_t . The family perceives an inflated current growth,

$$\tilde{x}_t = x_t + \zeta \varphi_e \bar{\sigma} \eta_t, \quad \zeta \geq 0, \quad (49)$$

which overweights the current innovation by the diagnostic parameter ζ , and it forecasts future growth with the *correct* persistence,

$$\mathbb{F}_t[x_{t+h}] = \rho^h \tilde{x}_t. \quad (50)$$

The two margins are separate. The diagnostic distortion ζ inflates the response of the perceived state to news and generates the overreaction: a positive surprise makes $\tilde{x}_t$ overshoot x_t , so the family’s forecast is systematically reversed by the subsequent data, the Coibion-Gorodnichenko

pattern (Section 7.10). Because the family uses the correct persistence, it expects the elevated growth, and the high value of hiring that comes with it, to fade at the true rate. This is why the family attributes a rise in valuations to near-term cash flows rather than to a permanent shift, which is what the subjective decomposition picks up (Section 7.12, Proposition 3). Firm-level beliefs apply the same diagnostic distortion to the idiosyncratic growth state $\xi_{i,t}$ with a firm-specific ζ^ξ , disciplined by the larger firm-level overreaction. Constant-gain learning of a slowly moving mean, as in Nagel and Xu (2022a) and Malmendier and Nagel (2015), is the special case ζ small and the perceived persistence high; the diagnostic form (49) separates the two so that both survey moments can be hit.

Perceived constant returns are implied, not assumed. The family watches the history of growth but does not track the variance state σ_t^2 ; it treats variance as fixed at $\bar{\sigma}^2$. Since the risk premium moves only with this variance, and the family does not see it, the family perceives a constant required return on hiring,

$$\mathbb{F}_t[R_{t+1}] = \bar{R} = r^f + \mathcal{K} \bar{\sigma}^2, \quad (51)$$

where $\mathcal{K}$ is the price-of-risk-times-exposure constant of Section 7.11. This is the pricing implication of the information set, not an additional restriction, and it removes the potential inconsistency between a family that prices with a kernel and one that perceives a constant return: under $\mathbb{F}$ the kernel's premium term is a constant, so the family prices consistently with (51). What varies objectively, and what the family does not perceive, is the return predictability created by its own diagnostic overreaction, since a high value of hiring is followed by low realized returns when the inflated growth $\tilde{x}_t$ reverts to x_t and x_t reverts at ρ (Proposition 2). This appears in the objective decomposition as discount-rate news and in the subjective decomposition as nothing, because the family attributes the valuation movement to its cash-flow forecasts.

The two measures. The subjective and objective measures agree on every object the family observes and differ only on the diagnostic distortion and the unobserved variance state, so $\mathbb{F}$ is a coherent measure and the equilibrium is internally rational (Appendix A.19).

7.8 Aggregation

Firms are equal-sized and of measure zero, so the aggregate is the equal-weight integral of the firm block: the aggregate value of hiring loads only on the common belief $\tilde{x}_t$, the idiosyncratic

beliefs wash out by the law of large numbers, and the cross-section is the dispersion $\Lambda \text{SD}(\tilde{\xi}_{i,t})$ matched in Section 7.14 (Appendix A.12).

7.9 Equilibrium

Because the family forms expectations under a subjective measure that need not equal the objective one, the equilibrium is an internally rational expectations equilibrium in the sense of Adam and Marcet (2011): the family optimizes under a well-defined subjective probability measure, and the realized data are generated by its policies under the objective measure.

The formal recursive definition of the internally rational equilibrium is in Appendix A.13.

7.10 Solution and the empirical mapping of the belief

State space and solution. The asset block is solved in closed form (the Epstein-Zin kernel and Campbell-Shiller loadings of Appendix A.1, and the value of hiring (52)); the labor block is log-linearized and the economy simulated monthly under the objective law with decisions under the subjective law $\mathbb{F}$, following Adam and Marcet (2011). Appendix A.14 gives the state space and the solution algorithm.

Mapping the belief to the survey data. The belief is disciplined by the survey moments it generates. The diagnostic distortion ζ and the perceived persistence ρ are separate margins that map to two distinct survey facts, which is the property that lets the belief match both.

Proposition 1 (The two survey moments are controlled by separate margins). *Under the diagnostic belief (49)-(50), (i) the Coibion-Gorodnichenko slope of the forecast error on the forecast revision is a strictly decreasing function of the diagnostic strength ζ , negative for $\zeta > 0$ and independent of the perceived persistence; and (ii) the subjective cash-flow share is $1 - (\kappa_{1m}\rho)^H$, a function of the perceived persistence ρ and independent of ζ (Appendix A.3).*

So ζ is disciplined by the measured overreaction and ρ by the persistence of long-run growth forecasts, and the subjective cash-flow share then follows without a further free parameter. Estimated on annual growth rates with firm and time fixed effects to match the empirical specification, the aggregate slope pins $\zeta = 5.0$, at which the model slope is about -0.49 ; the firm diagnostic strength ζ^ξ is instead disciplined by the cross-sectional return dispersion (Section 7.13). What the diagnostic belief attains, and no over-extrapolation-of-persistence belief can, is the overreaction and the high subjective cash-flow share *jointly*, because part (ii) is invariant to ζ .

7.11 Asset pricing and the value of hiring

The equity premium. The premium the family perceives is a constant, with prices of risk fixed by (γ, ψ) and its level scaled by the pricing member's cash-flow exposure s_D (Section 7.3); it is calibrated to five percent rather than predicted (Appendix A.15). The mechanism that moves unemployment is not a time-varying premium but the belief: when the growth belief x_t falls, the long-duration value of hiring falls, vacancies fall, and unemployment rises.

Growth-stock duration and the value of hiring. The value of hiring is a long-duration claim on cash flows that grow with the belief, priced under $\mathbb{F}$ with the constant perceived required return $\bar{R}$ of (51). Iterating the Campbell-Shiller identity forward under the diagnostic forecast (50), the log price-earnings ratio of the value of hiring is

$$pe_t = \underbrace{-\frac{\bar{R} - \mu}{1 - \kappa_{1m}}}_{\equiv \bar{p}\bar{e} \text{ (constant discount)}} + \Lambda \tilde{x}_t, \quad \Lambda = \frac{\kappa_{1m}\rho}{1 - \kappa_{1m}\rho}, \quad (52)$$

where κ_{1m} is the Campbell-Shiller constant of the dividend claim and $\bar{p}\bar{e}$ is the constant level implied by $\bar{R}$ and mean growth μ . The loading Λ uses the *true* persistence ρ , the growth-stock duration of a correctly-perceived long-lived claim; the excess volatility of valuations comes not from an inflated loading but from the diagnostic state $\tilde{x}_t$ (49), whose standard deviation exceeds that of x_t by a factor that rises with ζ . A by-product is that the loading is well away from the Campbell-Shiller singularity: with the anchored $\rho = 0.95$, $\kappa_{1m}\rho \approx 0.94$ and $\Lambda \approx 16$, whereas an over-extrapolation belief that generated comparable volatility through the loading alone would require $\rho_F \rightarrow 1$ and $\kappa_{1m}\rho_F \rightarrow 1$, putting a small number in the denominator and making the log-linear approximation fragile. The diagnostic form sources the volatility from the state, not the loading, so the linearization is accurate and the loadings are insensitive to κ_{1m} over its plausible range. The discount term enters the level, not the variation: with a constant perceived required return, the only source of *variation* in pe_t is the belief $\tilde{x}_t$. Under rational expectations $\zeta = 0$, $\tilde{x}_t = x_t$, and the value of hiring loads on the fundamental with the same Λ but without the diagnostic amplification, so its volatility falls; the belief is the source of the excess volatility. Objectively, the inflated growth $\tilde{x}_t$ reverts to x_t , which reverts at ρ , so a high pe_t is followed by low realized returns; the movement in pe_t is objectively discount-rate news (Proposition 2), even though it is priced by the family as constant-discount cash-flow news. This belief-induced return predictability, not a separate time-varying discount state, is the objective discount-rate variation in this economy.

7.12 The two decompositions

The filling rate q_t and the value of hiring can be decomposed into the news that moves them, once objectively (using realized ex-post returns and cash flows) and once subjectively (using the family's forecasts). The objective decomposition is the empirical fact that valuations move on discount rates; the subjective decomposition is the survey fact that forecasters attribute the movement to cash flows.

Proposition 2 (Objective discount share from the diagnostic distortion). *Suppose the value of hiring is priced on the diagnostic state $\tilde{x}_t = x_t + \zeta\varphi_e\bar{\sigma}\eta_t$ (49), while cash flows are realized under the objective law with persistence ρ . Then a Campbell-Shiller decomposition of the realized value of hiring at horizon H attributes to expected returns a share that is increasing in the diagnostic strength ζ , because the transitory component $\zeta\varphi_e\bar{\sigma}\eta_t$ of pe_t is entirely reversed by realized returns while contributing nothing to realized cash-flow news. At $\zeta = 0$ the share is the fundamental one, about a fifth at five years; it rises with ζ and, at the calibrated ζ , equals about seventy percent, matching the data. The diagnostic overreaction is the source of the objective discount-rate variation, since the family prices with a constant premium.*

Proposition 3 (Subjective cash-flow share under the anchored terminal). *Suppose the family perceives constant returns and forecasts growth with the true persistence ρ (the anchored terminal of (50)). Then the subjective decomposition of the value of hiring at horizon H attributes to cash flows the share*

$$\text{subjective cash-flow share} = 1 - (\kappa_{1m}\rho)^H, \quad (53)$$

and a share $(\kappa_{1m}\rho)^H$ to the perceived terminal; the perceived discount share is zero. Crucially, this share depends on the perceived persistence ρ , not on the diagnostic strength ζ : because the family anchors the long run to the truth, the subjective cash-flow share is invariant to how strongly the belief overreacts in the short run. This is what decouples the subjective cash-flow share from the Coibion-Gorodnichenko overreaction, which is governed by ζ (Proposition 1).

The two propositions are what resolve the central tension of the belief literature. A single over-extrapolation-of-persistence belief faces a bind: amplifying valuations and generating overreaction both call for a high perceived persistence, while a high subjective cash-flow share, by (53), calls for a low one so the family expects reversion within the horizon. The diagnostic belief breaks the bind by putting the two demands on two different parameters. The diagnostic strength ζ generates the overreaction and the objective discount share and amplifies the value of

Table 1: Calibration (monthly)

Parameter	Symbol	Value	Source / target
<i>Preferences and long-run risk</i>			
Risk aversion	γ	10	Bansal and Yaron (2004)
Elasticity of substitution	ψ	1.5	Bansal and Yaron (2004)
Expected-growth persistence (true)	ρ	0.95	$1 - \nu$ (coincides with Bansal and Yaron (2004))
Growth-shock loading	φ_e	0.053	long-run-risk process
Mean conditional vol	$\bar{\sigma}$	0.0028/mo	consumption growth vol
Vol-of-vol (objective, unobserved by family)	σ_w	9.5×10^{-6}	consumption vol-of-vol
Limited-participation share	s_D	0.12	equity premium (%)
<i>Beliefs</i>			
Diagnostic strength (aggregate)	ζ	5.0	CG overreaction / objective discount
Fading-memory gain	ν	0.05	$1 - \text{LTG-forecast persistence, est. } 0.051 \text{ (s.e. } 0.016)$
Diagnostic strength (firm)	ζ^ξ	0.5	firm return dispersion $\text{SD}_i(r)$
<i>Labor and wages</i>			
Job-finding rate	f	0.45/mo	average monthly finding rate
Separation rate	$\bar{\delta}$	0.028/mo	Kehoe et al. (2023), annual 0.286
Matching elasticity	α	0.5	Petrongolo and Pissarides (2001)
Worker bargaining weight	η_w	0.097	measured wage cyclicality ω
Profit share	$\bar{s}h$	0.088	Compustat, median <code>oiadp/sale</code>
Home production	b	0.40	flow value of non-work
Vacancy cost	κ	0.10	steady-state tightness

Notes: Monthly calibration. Preference and long-run-risk parameters follow the standard values of Bansal and Yaron (2004); the matching elasticity is the midpoint of the range surveyed by Petrongolo and Pissarides (2001); the annual separation rate of 0.286 is from Kehoe et al. (2023). The three externally estimated parameters are the fading-memory gain ν (from the persistence of long-term growth forecasts), the wage cyclicality $\omega = \eta_w$ (from the ratio of expected to realized wage cyclicality in the Survey of Consumer Expectations, the CFO survey, and the Livingston survey), and the profit share $\bar{s}h$ (the median ratio of operating income after depreciation to sales, `oiadp/sale`, over 235,760 Compustat firm-years, 1980–2022). The diagnostic strengths ζ and ζ^ξ and the participation share s_D are each set to a single belief or pricing moment, as described in the text.

hiring (Propositions 2 and 1), while the anchored persistence ρ delivers the high subjective cash-flow share (Proposition 3) and is invariant to ζ . One belief, disciplined by two survey moments, matches both. Proofs are in Appendix A.2.

7.13 Calibration

The model is calibrated at the monthly frequency. Table 1 lists every parameter, its value, and its source. Parameters fall into three groups: preferences and the long-run-risk process, taken from the asset-pricing literature; the belief parameters, disciplined by the survey evidence; and the labor and wage parameters, disciplined by the flow data and the survey wage.

Parameter choices. Each parameter’s value and source are stated in Table 1 and justified in detail in Appendix A.9; the mapping between parameters and the moments they are set to is summarized next.

What is targeted and what is not. Table 2 states the mapping transparently. Three parameters are estimated externally, from data other than the model’s moments: the fading-memory gain ν from the persistence of long-term growth forecasts, the wage cyclicality $\omega = \eta_w$ from survey wage expectations, and the profit share $\overline{sh}$ from the Compustat operating margin. The specifications and sources are in Appendix A.8. The preference and flow parameters are from the literature. The internal parameters are each set to one belief or pricing moment: s_D to the premium, the aggregate diagnostic strength ζ to the aggregate Coibion-Gorodnichenko overreaction, the firm diagnostic strength ζ^ξ to the cross-sectional return dispersion $SD_i(r)$, and $(\rho, \bar{\sigma})$ to consumption-growth volatility and the long-run-forecast persistence. Everything else in Table 3, including the level of unemployment volatility, the return and price-earnings volatilities, both the objective and the subjective decomposition shares, the autocorrelation and Beveridge correlation of unemployment, and the cross-section, is an over-identifying check. Two features of this design carry the weight of the exercise: because the profit share is measured rather than assumed, the level of unemployment volatility is a prediction; and because the belief is set to the overreaction, both decomposition shares are predictions rather than targets.

Table 2: Targets versus over-identifying predictions

Set to one model moment (target)	Over-identifying prediction (not targeted)
$s_D \rightarrow$ equity premium (5%)	$SD(U)$ and its two-belief decomposition
$\zeta \rightarrow$ aggregate CG overreaction; $\zeta^\xi \rightarrow SD_i(r)$	return volatility, price-earnings volatility
$(\rho, \bar{\sigma}) \rightarrow$ consumption vol / forecast persistence	$AC(U)$, Beveridge correlation
	objective discount share and horizon profile
	subjective cash-flow share (aggregate and firm)
	cross-section $SD_i(pe)$, $SD_i(hl)$
	the rational-expectations contrast (Table 4)

Notes: Externally measured, not fit to any model moment: the fading-memory gain ν (long-term-growth-forecast persistence), the wage cyclicality $\omega = \eta_w$ (survey wage expectations), the profit share $\overline{sh}$ (Compustat operating margin), and the preference and flow parameters (literature). Because the profit share is measured, unemployment volatility moves to the prediction column.

7.14 Quantitative results

Table 3 reports the model against the data, together with the rational-expectations version of the same economy. The rational column turns off both belief channels at their rational values:

Table 3: Model versus Data at the Five-Year Horizon

	Data	Model	Model (RE)
<i>Aggregate</i>			
equity premium $-\text{Cov}(m, r)$ (%)	5.0	5.0	0.9
risk-free rate (%)	1.5	1.5	1.5
risk-free-rate volatility (%)	0.9	0.6	0.6
Sharpe ratio	0.31	0.31	0.09
return volatility (%)	16.0	16.2	9.7
log price-earnings volatility	0.14	0.13	0.05
objective decomposition (discount share, %)	69	69	20
subjective cash-flow share (%)	97	97	80
$\text{SD}(U_t) \times 100$	2.10	1.86	0.1
$\text{AC}(U_t)$	0.94	0.93	0.78
Beveridge correlation	-0.89	-0.82	-0.87
<i>Cross section</i>			
$\text{SD}_i(pe_{i,t}) \times 100$	22.6	22.6	9.1
$\text{SD}_i(hl_{i,t}) \times 100$	15.70	15.9	7.6
$\text{SD}_i(r_{i,t}) \times 100$	5.7	5.9	3.8
subjective cash-flow share (%)	79	73	80

Notes: The Model column is the belief economy; the Model (RE) column is the same economy with correct beliefs and, because the expected outside option then tracks realized wages, a flexible wage. Under rational expectations the subjective and objective decompositions coincide, so the belief wedge, the gap between the 97 subjective cash-flow share and the 31 objective, vanishes: both attribute about 80 percent to cash flows and about 20 to discount rates. The equity premium, the risk-free rate, and the Sharpe ratio are standard postwar asset-pricing targets; the risk-free-rate volatility target refers to the smooth ex-ante real rate. Aggregate moments are simulated at the monthly frequency and aggregated to the year; the cross-section is a panel of firms with idiosyncratic beliefs. $pe_{i,t}$ is the firm log price-earnings ratio, $hl_{i,t}$ the firm log hiring rate, and $r_{i,t}$ the firm return.

the family holds correct beliefs about growth ($\zeta = 0$, so $\tilde{x}_t = x_t$) and about the required return, and because the worker's expected outside option then tracks realized wages ($\beta^w = 1$) the wage is flexible ($\omega = 1$).

The two belief distortions and their roles. The model contains two distinct, separately measured belief distortions with a clear division of labor. The diagnostic *growth* belief (49), measured from analyst cash-flow forecasts, prices the value of hiring: it generates the equity premium, the volatility of valuations, and both decomposition shares. The acyclical *wage-expectation* belief (Lemma 2), measured from survey wage expectations, generates the volatility of unemployment: with a rigid wage, the productivity-wage gap opens and operating leverage turns fundamental productivity movements into large swings in the value of a filled job. Both beliefs operate through the same object, the value of a filled job, which is at once a long-duration asset and the driver of job creation; that shared object is what ties the asset market to the labor market. The division of

labor is one of degree, not of separate blocks: the growth belief carries the decomposition shares and about half the volatility of valuations but only about a twentieth of the unemployment volatility, while the wage-expectation belief carries essentially all of the unemployment volatility and, through the same operating leverage, roughly the other half of the volatility of valuations.

Decomposing the unemployment volatility. With the wage cyclical and the profit share both measured (Section 7.13), the *level* $SD(U) = 1.86$ is a prediction, about eighty-eight percent of the data’s 2.10; no labor parameter is set to it. Turning off one belief at a time (Table 4) shows which does the work. Turning off the growth belief ($\zeta = 0$) while keeping the rigid wage lowers $SD(U)$ from 1.86 to 1.77, a reduction of about five percent, because unemployment responds to the value of a filled job, whose volatility is dominated by the operating-leverage productivity-wage gap rather than the forward growth belief. Turning off the wage-expectation belief alone (flexible wage, $\beta^w = 1$) collapses $SD(U)$ to 0.12, near the Shimer benchmark. The wage-expectation belief and the operating leverage together carry essentially all of the labor amplification, and the growth belief about a twentieth. Measured belief distortions thus drive unemployment fluctuations, chiefly through the acyclical wage-expectation belief, while the growth belief’s first-order role is in asset prices and the two decompositions.

Table 4: Unemployment volatility, one belief channel at a time

Economy	$SD(U_t) \times 100$
Both beliefs (baseline, predicted)	1.86
Growth belief off ($\zeta = 0$), wage still rigid	1.77
Wage flexible ($\beta^w = 1$), growth belief on	0.12
Both off (full rational)	0.10

Notes: Each row re-solves the model with one belief distortion removed, holding all parameters at their calibrated values. Removing the growth belief ($\zeta = 0$) with the wage still rigid lowers $SD(U)$ from 1.86 to 1.77, a five percent reduction; making the wage flexible ($\beta^w = 1$) with the growth belief still on lowers it to 0.12. The two channels are not additive because the operating-leverage amplification interacts with both. The wage-expectation belief and the operating leverage carry essentially all of the labor amplification; the growth belief accounts for about a twentieth of it, and its first-order role is in the asset-pricing block and the two decompositions.

The belief drives asset prices. The same growth belief generates the equity premium and the volatility of valuations. In the belief economy the premium is five percent and the log price-earnings ratio has a volatility of 0.13, close to the data; in the rational economy the premium falls to 0.9 percent and the volatility to 0.05. The belief is priced because, under limited participation, it moves the marginal investor’s consumption through the dividend. The return volatility of 16.2

percent matches the data and falls to 9.7 under rational expectations.

The two decompositions. The objective decomposition attributes about 69 percent of the variance of the value of hiring to discount-rate news, matching the data. This share is the output of Proposition 2: the diagnostic overreaction moves valuations that realized cash flows do not validate, and the reversal shows up objectively as expected-return variation. Its level is governed by the diagnostic strength ζ , which is disciplined by the overreaction, so the objective share is not a free target. The subjective decomposition attributes about 97 percent at the aggregate and 73 in the cross-section to cash flows, close to the 97 and 79 in the survey data, because the family perceives constant returns and, anchoring the long run to the truth, expects the valuation to revert within the horizon (Proposition 3). The two decompositions are reproduced *jointly* by one belief: the diagnostic strength sets the objective share and the overreaction, while the anchored persistence sets the subjective share, and because these are separate margins (Proposition 1) the model no longer faces the tension between them that defeats a pure over-extrapolation belief. Under rational expectations the two decompositions coincide, both attributing about 80 percent to cash flows, so the belief wedge between the 97 subjective and the 31 objective cash-flow share vanishes.

Two over-identifying moments miss the data. The firm subjective cash-flow share is 73 against 79, a six-point understatement that follows from the anchored firm persistence and is not separately tunable; and the Beveridge correlation, -0.82 against -0.89 , is slightly weaker under the belief than under rational expectations (-0.87), so on this one dimension the distortion moves the model marginally away from the data. Neither is central, and both are reported rather than suppressed.

Overidentification and return predictability. Two further checks, that the objective discount share, and the return-predictability and belief-sorted-spread patterns, come out as the model implies without being targeted, are reported in Appendix A.10.

8 Robustness Checks and Extensions

This section presents additional results that reinforce the main finding that belief distortions about expected cash flows explain a large share of firms' hiring decisions.

Managerial vs. Analyst Expectations The paper primarily uses IBES analyst forecasts; as a robustness check, this section incorporates one-year-ahead earnings forecasts from the CFO

Survey, which reflect managerial expectations (Gennaioli et al., 2016; Hillenbrand and McCarthy, 2024). Many CFOs focus on pro-forma earnings, which exclude special items and are conceptually close to the “Street earnings” analysts forecast in IBES, so the two likely target similar concepts. Consistent with this interpretation, the two survey measures have a positive correlation of 0.60 at the one-year horizon. Moreover, Appendix Table A.1 shows that replacing IBES expectations with CFO expectations yields similar variance decomposition results: cash flow expectations remain the dominant driver of hiring fluctuations under subjective managerial beliefs implied by a survey of CFOs.

Financial Constraints Figure A.4 shows that controlling for five firm-level financial constraint proxies (firm size, payout ratio, SA index, expected free cash flow, and Whited-Wu index) only modestly reduces the contribution of expected earnings under subjective expectations but leaves most of the decomposition intact. Under objective expectations, however, discount rate contributions drop substantially, consistent with rational models in which tightening financial constraints can influence the discount rate.

9 Conclusion

This paper develops a framework linking asset valuations and hiring to distortions in firms’ beliefs about future cash flows, explaining both aggregate fluctuations and cross-sectional dispersion. Using machine learning forecasts as an objective benchmark, I document that survey forecasts systematically overreact to cash flow news while machine forecasts do not. These distortions make hiring highly sensitive to perceived cash flow news, explaining over 60% of variation in both aggregate and firm-level hiring. Firms with distorted beliefs overhire after positive idiosyncratic shocks, reducing realized profits per worker, while objective firms do not.

A general-equilibrium search-and-matching model with Epstein-Zin preferences and limited asset-market participation reproduces these patterns from two separately measured belief distortions. A diagnostic growth belief overreacts to recent cash-flow news while anchoring the long run to the truth; priced by the household’s stochastic discount factor, it generates the equity premium, the volatility of valuations, and the value the firm places on a new hire. An acyclical belief about the expected re-employment wage, read from survey wage expectations, makes the wage rigid, so that operating leverage turns productivity movements into large swings in the value of a filled job and hence in vacancies and unemployment. Both distortions act through the same object, the value of hiring, which is at once a long-duration asset and the driver of job creation, so asset prices and hiring move together. The model reproduces the objective decomposition

of valuations onto discount rates and the subjective decomposition onto cash flows jointly, and predicts an unemployment volatility about 88% of that in the data, with no labor parameter set to match it. Under rational expectations both distortions vanish, the wage becomes flexible, and unemployment volatility collapses toward the frictionless benchmark, which is the quantitative content of the belief distortions.

Belief distortions offer a common lens on business cycles, linking asset valuations and real activity. Because hiring is a forward-looking decision valued like any other asset, the biases that drive asset valuations can also drive hiring, capital investment, R&D, and broader business-cycle dynamics. Incorporating belief formation into macroeconomic models is thus a promising avenue for future research.

The findings also bear on stabilization policy. Because firms overreact to economic conditions, monetary policy can lean against the wind to temper optimism, though the same overreaction calls for cautious, state-contingent calibration to avoid amplifying volatility. Unemployment insurance may need to be more generous in recessions, when pessimistic overreaction prolongs unemployment beyond what fundamentals justify. Forward guidance can help anchor expectations when uncertainty is high. Overreaction to idiosyncratic shocks also generates excess dispersion in hiring, raising the variance of marginal products across firms and lowering aggregate productivity through misallocation. Finally, measured belief distortions can serve as an early-warning indicator for shifts in unemployment and labor demand.

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A The General-Equilibrium Model of Section 7: Derivations and Proofs

A.1 The Epstein-Zin long-run-risk kernel

The kernel (37) is solved in closed form by the Campbell-Shiller log-linearization of the Bansal-Yaron (2004) economy. Let z_t be the log wealth-consumption ratio of the pricing member and κ_0, κ_1 the linearization constants at $\bar{z}$. The consumption claim $z_t = A_0 + A_1 x_t + A_2 \sigma_t^2$ has loadings

$$A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}, \quad A_2 = \frac{\frac{1}{2}[(\theta_{EZ} \kappa_1 A_1 \varphi_e)^2 + (1 - \gamma)^2]}{\theta_{EZ}(1 - \kappa_1 \nu_\sigma)},$$

with $A_1 > 0$ for $\psi > 1$ and $A_2 < 0$ (uncertainty lowers the price). The dividend claim $z_t^m = A_{0m} + A_{1m} x_t + A_{2m} \sigma_t^2$ has $A_{1m} = (\phi_D - 1/\psi)/(1 - \kappa_{1m} \rho)$, with dividend leverage $\phi_D > 1/\psi$. The log kernel innovation is (A.10) with prices of risk (A.11). The equity premium is

$$\mathbb{E}_t[r^e - r^f] = (1 - \theta_{EZ})\kappa_1 \kappa_{1m} A_1 A_{1m} \varphi_e^2 \sigma_t^2 + (1 - \theta_{EZ})\kappa_1 \kappa_{1m} A_2 A_{2m} \sigma_w^2,$$

the first term time-varying through σ_t^2 and the second a constant variance-risk premium. The solver reproduces the published Bansal-Yaron (2004) moments as a validation gate: equity premium 6.84%, risk-free rate 1.34%, return volatility 18.9%, Sharpe ratio 0.36; the code returns 6.8%, 1.3%, 18.9%, 0.36. Under limited participation the pricing member's consumption growth loads on the dividend with share s_D , so A_1 and the prices of risk are scaled by the exposure of C^p to x_t ; setting $s_D = 0.12$ delivers the calibrated premium.

A.2 Proofs

Proof of Lemma 1. The dividend is $D = Y - W - F$ with F fixed, so $dD = dY - dW$. Dividing by the steady-state level $D = \bar{s}\bar{h}Y$ and writing deviations $\hat{X} = dX/X$, $\hat{D} = (Y/D)\hat{Y} - (W/D)\hat{W} = \bar{s}\bar{h}^{-1}\hat{Y} - (sh_W/\bar{s}\bar{h})\hat{W}$, where $sh_W = W/Y$. The elasticity of the dividend to output, holding the wage, is $1/\bar{s}\bar{h}$, the degree of operating leverage; the fixed cost F has $dF = 0$ but enters the level through $\bar{s}\bar{h} = 1 - sh_W - sh_F$, so a larger fixed-cost share sh_F lowers $\bar{s}\bar{h}$ and raises the leverage. With only labor costs ($sh_F = 0$) the margin would be $\bar{s}\bar{h} = 1 - sh_W$, the complement of the labor share, and the leverage small; the fixed cost is what reconciles the high leverage the earnings volatility requires with the small operating margin firms report. ■

Proof of Lemma 2. The cyclicalty is measured relative to the stochastic trend e^{at} , the object the empirical wage-cyclicalty regressions detrend by. Write $z_t^w = b e^{at} + z_t^{w,f}$, where $b e^{at}$ is home production and $z_t^{w,f}$ is the forward-looking continuation of (45). Detrended by e^{at} , home production is the constant b and contributes no cyclicalty; the trend component of every term is common and differenced out. Log-linearizing the detrended (47), the marginal-product term contributes $\eta_w d \log A$ (the deviation of productivity from its own trend), the tightness term is second order at the deterministic steady state, and the outside-option term contributes $(1 - \eta_w) d \log z^{w,f}$, whose elasticity to the productivity deviation is β^w by definition. Collecting terms, $d \log W = [\eta_w + (1 - \eta_w)\beta^w] d \log A$, which is (48); the unit-elastic home-production term enters the trend, not the cyclicalty, and $\beta^w \approx 0$ is the acyclicalty of the detrended forward-looking outside option, the object the surveys measure. ■

Proof of Proposition 3. Under perceived constant returns the family's Campbell-Shiller identity for the value of hiring is $pe_t = \sum_{h \geq 1} \kappa_{1m}^{h-1} \mathbb{F}_t[\Delta d_{t+h}] - \text{const}$, since the perceived discount term is constant. The family forecasts growth with the anchored persistence ρ (50), so $\mathbb{F}_t[\Delta d_{t+h}] = \rho^{h-1} \Lambda^{-1} pe_t$ up to the loading, and the H -horizon cash-flow term is $pe_t \sum_{h=1}^H (\kappa_{1m} \rho)^{h-1} (1 - \kappa_{1m} \rho) = pe_t [1 - (\kappa_{1m} \rho)^H]$, while the perceived terminal is $\kappa_{1m}^H \mathbb{F}_t[pe_{t+H}] = (\kappa_{1m} \rho)^H pe_t$. The two sum to pe_t , so the cash-flow share is $1 - (\kappa_{1m} \rho)^H$ and the terminal share is $(\kappa_{1m} \rho)^H$, with zero perceived discount share. Both depend on ρ and not on the diagnostic strength ζ , since ζ enters only the level of $\tilde{x}_t$ and cancels in the shares. ■

Proof of Proposition 2 (sketch). The realized return is $r_{t+1} = \kappa_{1m}pe_{t+1} - pe_t + \Delta d_{t+1}$ with $pe_t = \Lambda \tilde{x}_t$ and $\tilde{x}_t = x_t + \zeta \varphi_e \bar{\sigma} \eta_t$. The diagnostic component $\zeta \varphi_e \bar{\sigma} \eta_t$ is transitory: it is in pe_t but not in the expected next-period price, which is $\kappa_{1m} \Lambda \rho x_t$ under the objective law, so it is fully reversed by the realized return and contributes nothing to realized cash-flow news. The fundamental component x_t reverts at ρ and contributes the fundamental discount share. Projecting the realized H -horizon return sum on pe_t , the discount share is a weighted average of the fundamental share and one (the diagnostic component is pure discount-rate news), with weight on the latter equal to the variance share of the diagnostic component, $\zeta^2/(\zeta^2 + \text{fundamental})$, which rises in ζ . At the calibrated $\zeta = 5$ and $H = 60$ months it evaluates to about seventy percent, matching the data; at $\zeta = 0$ it is the fundamental share of about a fifth. Since the family prices with a constant premium, this belief reversal is the entire objective discount-rate variation. ■

A.3 Proof of Proposition 1: the two survey margins

Part (ii), the subjective share. Shown in the proof of Proposition 3 above: the share is $1 - (\kappa_{1m}\rho)^H$, a function of the anchored persistence ρ alone, because the diagnostic term enters $\tilde{x}_t$ multiplicatively in the level and cancels in the shares.

Part (i), the CG slope. Under the diagnostic forecast (50) the H -horizon cumulative-growth forecast is $F_t = c \tilde{x}_t$ with $c = \sum_{h=1}^H \rho^h$ and $\tilde{x}_t = x_t + \zeta u_t$, $u_t = \varphi_e \bar{\sigma} \eta_t$ the current innovation. The revision is $\Delta F_t = c(\Delta x_t + \zeta \Delta u_t)$. The realized cumulative growth has conditional mean $c x_t$, so the forecast error is $-c \zeta u_t$ plus terms orthogonal to the revision. Writing $\sigma_u^2 = \text{Var}(u_t)$ and using $\text{Cov}(x_t, u_t) = \sigma_u^2$, $\text{Cov}(\Delta x_t, u_t) = \sigma_u^2$, and $\text{Var}(\Delta u_t) = 2\sigma_u^2$,

$$\beta_{CG} = \frac{\text{Cov}(-c \zeta u_t, c(\Delta x_t + \zeta \Delta u_t))}{\text{Var}(c(\Delta x_t + \zeta \Delta u_t))} = \frac{-\zeta [\sigma_u^2 + \zeta \sigma_u^2]}{2(1 - \rho)\text{Var}(x_t) + 2\zeta \sigma_u^2 + \dots},$$

which is zero at $\zeta = 0$ and strictly decreasing in ζ , so the slope is negative for $\zeta > 0$, with a sign independent of the anchored persistence (its magnitude varies mildly with ρ). This establishes part (i). What the diagnostic form adds is that it delivers its overreaction while part (ii) holds the subjective share fixed at the anchored value, so the two survey facts no longer conflict.

A.4 One cognitive family, two beliefs

This appendix formalizes the claim of Section 7.7 that the growth belief and the wage belief are two manifestations of one behavioral disposition, and shows why a single parameter cannot replace the two.

The two operators. The disposition is: over-weight the salient aggregate innovation, and anchor the own status-quo position. Applied to the aggregate growth state it is the diagnostic operator (49), $\tilde{x}_t = x_t + \zeta u_t$ with $u_t = \varphi_e \bar{\sigma} \eta_t$, which overweights the current innovation by $\zeta \geq 0$. Applied to the worker's own re-employment wage it is a reference-dependent operator that mixes a slow-moving status-quo norm $\bar{W}_t$ with the rational forecast,

$$\mathbb{F}_t^w[W_{t+1}^{\text{new}}] = \phi \bar{W}_t + (1 - \phi) \mathbb{E}_t[W_{t+1}^{\text{new}}], \quad \phi \in [0, 1], \quad (\text{A.1})$$

where $\bar{W}_t$ does not respond to the current cycle. Log-linearizing, the cyclicity of the expected outside option is $\beta^w = 1 - \phi$, so the wage elasticity of Lemma 2 is $\omega = \eta_w + (1 - \eta_w)(1 - \phi)$. Full anchoring $\phi \rightarrow 1$ gives $\beta^w \rightarrow 0$ and the rigid wage; $\phi = 0$ gives rational expectations and the flexible wage.

The signs are opposite, so one parameter cannot serve. The aggregate operator over-extrapolates the cycle and the own-wage operator under-extrapolates it. Suppose instead the worker forecast the own wage with the *same* diagnostic operator that prices the aggregate, ζ on the current innovation. The outside option is a long-duration present value of the perceived re-employment wage, which tracks perceived productivity; under the diagnostic operator its cyclicity is the value-of-hiring loading applied to an over-weighted innovation, which exceeds unity. Numerically, at the calibrated $\zeta = 5$ the shared-belief outside option has $\beta^w = 1.39$, hence $\omega = 1.35$: the wage would be *more* than fully flexible, and unemployment volatility would collapse to $\text{SD}(U) = 0.14$, below the rational-expectations benchmark, rather than reproducing the data. The intuition is direct: a worker who over-perceives aggregate growth also over-perceives the value of the next job, so the threat point rises in booms and the wage becomes more procyclical. Reproducing the rigidity therefore requires the own-wage operator to carry the opposite sign, ϕ large rather than ζ large, which is what the wage surveys show.

Joint discipline. The two parameters are set by two independent surveys and are not interchangeable: $\zeta = 5$ from the Coibion-Gorodnichenko overreaction in analyst cash-flow forecasts (Appendix A.3, model slope -0.49), and $\phi = 0.91$ from the acyclicity of survey wage expectations ($\beta^w = 1 - \phi = 0.09$, within the measured $[0.05, 0.10]$). At this pair the model reproduces both survey slopes and, as an over-identifying check, an unemployment volatility of the right order that no labor parameter targets. A single parameter, whether a shared diagnostic strength or a shared anchoring weight, cannot hit both survey slopes at once, because they have opposite signs.

A.5 The monthly labor block

Solved directly at the annual frequency the flow equation (43) is degenerate: the monthly job-finding rate $f = 0.45$ implies the unemployment pool turns over within the year, and the annual matching normalization yields a steady-state unemployment rate near one third and an unstable annual law. The block is therefore solved monthly and aggregated. In the two-margin version the separation rate is $\delta_t = \bar{\delta} + \phi_\delta \widehat{(-P_t)}$, countercyclical in the value of hiring with elasticity ϕ_δ set to the one-quarter separation share of Shimer (2012). The monthly unemployment deviation follows the linearized law $dU_t = \rho_u dU_{t-1} - k_{resp} \hat{P}_t$ with $\rho_u = 1 - \bar{\delta}_m - f_m$ and $k_{resp} = (1 - \bar{U})\bar{\delta}_m\phi_\delta + \bar{U}f_me_f$, where e_f is the free-entry elasticity of job finding to the value of hiring. Annual unemployment is the end-of-year value and its standard deviation and autocorrelation are computed on the annual series.

A.6 Limited participation: the participation cutoff

This appendix derives the reduced-form share $s_D = \omega_D/\Pi$ of Section 7.3 from the participation problem and shows how it scales the prices of risk. Each member of the family draws a fixed per-period utility cost $\varkappa$ of holding equity from a continuous distribution G on $[0, \infty)$. Let $\Delta(\varkappa)$ be the certainty-equivalent gain, in consumption units and evaluated under the family kernel, from holding the concentrated equity position net of the cost. A member participates if and only if $\varkappa \leq \bar{\varkappa}$, where $\bar{\varkappa}$ solves $\Delta(\bar{\varkappa}) = 0$, so the participating measure is $\Pi = G(\bar{\varkappa})$ and the marginal participant is indifferent. Participants hold the entire equity float, so a participant's equity position per unit of consumption is the aggregate float scaled by $1/\Pi$; to first order the dividend income of a participant per unit of *his* consumption is ω_D/Π , where $\omega_D = D/C$ is the aggregate payout ratio. The pricing member's consumption growth is therefore (38) with

$$s_D = \frac{\omega_D}{\Pi}. \quad (\text{A.2})$$

Two comparative statics follow. First, s_D falls with participation and rises with the payout ratio, both observable; at $\omega_D \approx 0.03$ and $\Pi \approx 0.25$, a quarter of households in line with the Survey of Consumer Finances, $s_D \approx 0.12$. Second, in the kernel the pricing member's consumption loads on the growth state x_t with the scaled loading $s_D A_{1m}$ in place of A_1 , so the prices of risk (A.11) and the premium (A.9) scale linearly in s_D : the growth belief supplies the covariance of the kernel with the value of hiring, and s_D sets its magnitude. At $s_D = 0$ the kernel prices only smooth aggregate consumption and the premium is 2.8 percent; at $s_D = 0.12$ it is five percent. This is the sense in which limited participation is load-bearing and disciplined rather than free: the belief does the covariance work and s_D , pinned by the participation rate and the payout ratio, scales it.

Consumption aggregation. The kernel prices with the participant's consumption C^p , but the resource constraint holds for aggregate consumption C . The two are consistent: aggregate consumption is the population average $C_t = \Pi C_t^{\text{part}} + (1 - \Pi)C_t^{\text{non}}$, where participants consume labor income plus dividends and non-participants consume labor income only, and goods-market clearing (Section 7.9) determines the level of C_t . The dividend appears once in C_t , in the participants' term; it is not double-counted. What the kernel uses is the growth of the marginal participant's consumption, (38), whose dividend loading s_D is the concentration $1/\Pi$ times the payout share, as above. Aggregate consumption growth is smooth because dividends are a small share of it; the participant's is volatile because dividends are a large share of *his*. Both statements hold simultaneously and are the content of limited participation.

A.7 The value of hiring and unemployment

The value of hiring is the forward-iterated Bellman claim

$$\hat{P}_{i,t} = \mathbb{E}_t^{\mathbb{F}} \sum_{k \geq 0} \left[\prod_{j=1}^k (1 - \delta_{t+j}) M_{t+1+j} \right] \hat{D}_{i,t+1+k}, \quad (\text{A.3})$$

whose loading on the growth belief is the growth-stock duration Λ of Section 7.11.

This appendix collects the free-entry and job-creation results that connect the value of hiring to unemployment. A vacancy costs κe^{a_t} per period and is filled with probability $q(\theta_t) = \chi \theta_t^{-\alpha}$, where $\theta_t = v_t/U_t$ is market tightness and α is the matching elasticity; the job-finding rate is $f(\theta_t) = \chi \theta_t^{1-\alpha}$. Free entry drives the value of a vacancy to zero, so from (41)

$$\kappa e^{a_t} = q(\theta_t) \mathbb{E}_t^{\mathbb{F}}[M_{t+1} J_{i,t+1}], \quad (\text{A.4})$$

the vacancy cost equals the filling probability times the expected discounted value of a filled job. Log-linearizing, the filling rate moves inversely with the value of hiring, $\hat{q}_t = -\hat{P}_t$ up to a constant, and because $q(\theta) = \chi \theta^{-\alpha}$, tightness responds with elasticity $1/\alpha$,

$$\hat{\theta}_t = \frac{1}{\alpha} \hat{P}_t, \quad \hat{f}_t = (1 - \alpha) \hat{\theta}_t = \frac{1-\alpha}{\alpha} \hat{P}_t \equiv e_f \hat{P}_t. \quad (\text{A.5})$$

The job-finding elasticity $e_f = (1 - \alpha)/\alpha$ is the coefficient in the unemployment law of Appendix A.5; it decreases in α , so a higher matching elasticity dampens the response of unemployment to the value of hiring. This is the source of the α -sensitivity band reported in Section 7.13: across the range $\alpha \in [0.4, 0.6]$ the prediction $\text{SD}(U)$ moves from about 2.2 to 1.6, with 1.86 at the midpoint $\alpha = 0.5$.

The value of hiring itself loads on both beliefs. From the Bellman equation (40), $J_{i,t} = e^{a_t} \pi_{i,t} + \mathbb{E}_t^{\mathbb{F}}[M_{t+1}(1 - \delta_{t+1})J_{i,t+1}]$ with $\pi_{i,t}$ the flow profit per unit of productivity. Dividing by productivity, substituting the operating-leverage relation (Lemma 1) for $\pi_{i,t}$ and the wage rule (47), and log-linearizing gives the log value of hiring as

$$\hat{P}_t = A \tilde{x}_t + B g_t + C \hat{L}_{t-1}, \quad (\text{A.6})$$

where g_t is the productivity-wage gap, $\hat{L}_{t-1}$ is lagged employment, and (A, B, C) are solved at the fixed point of the recursion in Appendix A.5. The growth belief enters through $\tilde{x}_t$ and the wage rigidity through g_t : the diagnostic distortion moves $\tilde{x}_t$, and the acyclical wage lets the productivity-wage gap g_t open, so both beliefs move $\hat{P}_t$ and, through free entry, vacancies and unemployment. Turning off the growth belief sets $\zeta = 0$ in $\tilde{x}_t$; turning off the wage rigidity sets $\beta^w = 1$, which closes g_t . Table 4 reports the effect of each.

A.8 Data and the measured parameters

Three parameters are estimated outside the model, and two belief facts discipline the diagnostic strength and the decompositions. This appendix states the specifications and sources.

Fading-memory gain ν and perceived persistence ρ . The perceived persistence is read from the persistence of long-term growth forecasts: regressing the smoothed long-run forecast on its own lag gives an autoregressive coefficient of about 0.95, consistent with the family anchoring the long run to the truth, and the fading-memory gain is $\nu = 1 - 0.95$, estimated at 0.051 with a standard error of 0.016. This is the constant-gain representation of Nagel and Xu (2022a) and Malmendier and Nagel (2015).

Wage cyclicity ω . The wage elasticity $\omega = \eta_w + (1 - \eta_w)\beta^w$ (Lemma 2) is measured from the cyclicity of expected relative to realized wage growth. Regressing expected wage growth on the change in unemployment gives slopes of +0.003 in the Survey of Consumer Expectations, +0.001 in the CFO survey, and +0.005 in the Livingston survey; regressing realized wage growth on the same change gives -0.048 and -0.062 . The ratio of expected to realized cyclicity is $\beta^w \in [0.05, 0.10]$, so with an acyclical outside option $\omega = \eta_w \in [0.05, 0.10]$, and the baseline sets $\eta_w = 0.097$.

Profit share $\overline{sh}$. The profit share is the median ratio of operating income after depreciation to sales (`oiadp/sale`) in the Compustat annual file, 0.088 across 235,760 firm-years over 1980–2022. By Lemma 1 the operating leverage is $L_{op} = 1/\overline{sh} = 11.4$. The mean margin (0.098) gives $L_{op} = 10.2$ and $SD(U) = 1.64$; the median is reported as the baseline.

The Coibion-Gorodnichenko overreaction. The diagnostic strength ζ is set to the overreaction in analyst earnings-growth forecasts, measured as the slope of the forecast error on the forecast revision, run on annual growth rates with firm and time fixed effects so that the time effect removes the common component and the firm coefficient reflects the idiosyncratic overreaction. The aggregate slope pins $\zeta = 5.0$, at which the model slope is -0.49 ; the firm slope is stronger, and the firm diagnostic strength ζ^ξ is instead identified by the cross-sectional return dispersion, as in the calibration.

The objective and subjective decompositions. The objective decomposition uses a machine forecast of returns and cash flows, trained by long short-term memory on the same information set, as the ex-ante conditional expectation, and attributes about 69 percent of the variance of the value of hiring at the five-year horizon to discount-rate news, following De La O et al. (2024). The subjective decomposition uses survey forecasts, CFO expected returns and analyst cash-flow forecasts, and attributes about 97 percent at the aggregate and 79 in the cross-section to cash flows. The wedge between the two, valuations that move on discount rates but that forecasters attribute to cash flows, is the belief distortion the model reproduces.

A.9 Calibration details

Preferences and long-run risk. Risk aversion $\gamma = 10$ and the elasticity $\psi = 1.5$ are the standard long-run-risk values; with $\psi > 1$ the wealth-consumption loading $A_1 > 0$ and the prices of risk (A.11) are positive. The true persistence $\rho = 0.95$ and the conditional volatility $\bar{\sigma}$ are set so that the implied consumption-growth volatility matches the data. The volatility state σ_t^2 is an objective feature the family does not condition on (Section 7.7); because the family prices with a constant perceived premium, σ_t^2 affects realized return and consumption volatility but not the decomposition of valuations, so the objective discount share is generated by the belief reversal (Proposition 2), not by a tuned stochastic-variance channel. The prices of risk are not free: they are the functions (A.11) of (γ, ψ) through the wealth-consumption loadings, and the same kernel prices the risk-free rate and the value of hiring. Appendix A.1 validates the closed-form solver against Bansal-Yaron (2004): premium 6.8%, risk-free rate 1.3%, return volatility 18.9%.

Limited participation. The dividend share $s_D = 0.12$ is the one parameter that translates the volatility of firm cash flows into the volatility of the pricing kernel. It is disciplined by the equity premium: with the endogenous prices of risk fixed by (γ, ψ) , $s_D = 0.12$ delivers a premium of five percent, and the endogenous consumption of the marginal investor loads on dividends with this share. Setting $s_D = 0$ prices only smooth aggregate consumption (growth 0.4% per year) and delivers a premium of 2.8%; the dividend share is the empirically standard observation that the marginal investor’s consumption is more exposed to equity than the representative household’s.

Beliefs. The perceived persistence is the true $\rho = 0.95$, estimated from the persistence of long-term growth forecasts: the smoothed long-run forecast has autoregressive coefficient about 0.95, consistent with the family anchoring the long run to the truth. The diagnostic strength ζ is the only free belief parameter and is disciplined by the measured overreaction. By Proposition 1 the Coibion-Gorodnichenko slope is strictly decreasing in ζ ; the aggregate slope pins $\zeta = 5.0$, at which the model slope is about -0.49 , and the objective discount share is then 69 percent, matching the data (Proposition 2). The subjective cash-flow share follows without a further parameter, $1 - (\kappa_{1m}\rho)^{60} \approx 0.97$, matching the survey. The firm diagnostic strength $\zeta^\xi = 0.5$ is set to the cross-sectional return dispersion $SD_i(r)$: the firm return in the model is dominated by the revision of the idiosyncratic value of hiring, so ζ^ξ is identified by $SD_i(r)$, and at $\zeta^\xi = 0.5$ the model matches the dispersion moments $SD_i(pe)$, $SD_i(hl)$, $SD_i(r)$ and the firm subjective cash-flow share.

Labor and wages. The job-finding and separation rates are the monthly counterparts of the annual flows; the matching elasticity $\alpha = 0.5$ is the mid-point of the Petrongolo-Pissarides range, and because the job-creation condition (42) gives a job-finding elasticity $e_f = (1 - \alpha)/\alpha$ that falls in α (Appendix A.7), the prediction moves within a band of roughly ± 15 percent across that range, from 1.6 at $\alpha = 0.6$ to 2.2 at $\alpha = 0.4$. The two parameters that govern the volatility of unemployment, the wage cyclicality and the operating leverage, are both measured, so that $SD(U)$ is a prediction rather than a target. The worker bargaining weight $\eta_w = 0.097$ is set to the measured wage cyclicality: with the expected outside option acyclical, Lemma 2 gives $\omega = \eta_w$, and the survey ratio of expected to realized wage cyclicality is 0.05 to 0.10. The profit share $\bar{s}h = 0.088$ is measured from the Compustat annual file: the median ratio of operating income to sales (operating income after depreciation, `oiadp/sale`) over 1980–2022 is 0.088 across 235,760 firm-years, which by Lemma 1 gives operating leverage $L_{op} = 1/\bar{s}h = 11.4$. Neither parameter is chosen to fit $SD(U)$. Given both, the model predicts $SD(U) = 1.86$ percentage points against 2.10 in the data, about eighty-eight percent, and the shortfall is reported as such. The prediction is sensitive to the profit share in the way Lemma 1 implies: the median operating margin gives 1.86, the mean margin (0.098) gives 1.64, and a marginal-cost surplus of five percent would over-predict at 3.6; the measured margin sits where the model captures the bulk of unemployment volatility without the small surplus of Hagedorn and Manovskii (2008) that the measured-wage approach is designed to avoid.

A.10 Additional quantitative results

Return predictability and the cross-section of returns. Two sharper asset-pricing predictions follow, both of the diagnostic mechanism and both testable. First, because the diagnostic overshoot reverses, a high price-earnings ratio predicts low future returns: regressing the H -year market return on pe_t , the model produces a negative slope that steepens with the horizon and an R^2 that rises from about 0.05 at one year to about 0.25 at five, in line with the return-predictability evidence of Cochrane (2011) and the objective discount share of Proposition 2. Second, in the cross-section the model implies a belief-sorted return spread: firms with a high idiosyncratic diagnostic belief $\tilde{\xi}_{i,t}$ are the high- pe , high-growth firms, and they earn low subsequent returns as their overshoot reverts. This is the mechanism of Bordalo et al. (2019), who document that high-long-term-growth-forecast firms underperform; the model reproduces the sign and locates the spread in the same object, the survey growth expectation, and predicts that the spread is largest for the firms whose CG overreaction is strongest. The cross-sectional dispersions of the price-earnings ratio, the hiring rate, and returns are generated by the dispersion of $\tilde{\xi}_{i,t}$ and collapse under rational expectations. The model’s constant *perceived* return is, in contrast, at odds with the procyclical survey *expected* returns of Greenwood and Shleifer (2014); the model speaks to subjective cash-flow expectations, which it matches, and takes the perceived constant return as the pricing counterpart of the family not conditioning on the volatility state, an assumption that a richer belief over returns would need to relax.

Overidentification. The objective discount share rises with the horizon, from about a quarter at one year to about sixty-nine percent at five, matching the data, a profile the model reproduces and does not target: no parameter is set to the objective discount at any horizon, so both the shape and the level are outputs of the diagnostic distortion (Proposition 2) whose strength is set to the overreaction. Likewise the subjective cash-flow share of 97 follows from the anchored persistence and is not targeted. The Beveridge correlation of -0.82 , the autocorrelation of unemployment of 0.93, the return volatility, and the price-earnings volatility are outcomes rather than targets. The calibrated parameters and the moments they are set to are listed in Section 7.13; the moments in this paragraph are the over-identifying checks.

A.11 Household structure: risk sharing, timing, and balanced growth

Risk sharing and market structure. The extent of risk sharing inside the family is the assumption that determines both the labor-supply block and the pricing kernel, and the two are treated differently. *Employment risk is fully shared:* the family provides each member complete insurance against the idiosyncratic risk of being employed or unemployed, so all members consume the same amount in each state and there are no wealth effects on the labor-supply and search margins. This is the Merz-Andolfatto device and it is what lets a single value function (36) represent the family. *Equity risk is not fully shared:* the firm equity is held and priced by a subset

of members, the stockholders, who cannot fully diversify the dividend stream against the non-stockholders, as in the limited-participation structure of Mankiw and Zeldes (1991) and Guvenen (2009). The marginal utility that prices equity is therefore the stockholders', and their consumption is more exposed to the dividend than average family consumption; the exposure is the share s_D of (38), micro-founded in Section 7.3. Two features follow. First, the labor market behaves as in a full-insurance search model, so the Merz-Andolfatto aggregation is exact for the employment block. Second, the asset market is segmented, so the kernel is not priced by smooth aggregate consumption; without this segmentation ($s_D = 0$) the belief about firm cash flows could not move the kernel and the premium would collapse to the smooth-consumption value. The two assumptions are not in tension: full insurance is over employment states, imperfect sharing is over the equity return, and they apply to different margins.

Timing, markets, and balanced growth. Within a period, the aggregate shocks $(\varepsilon_t, \eta_t, w_t)$ and the idiosyncratic shocks realize; matches form and separate according to (43); production, wage payment, and consumption take place; and equity is priced. Markets are complete within the stockholder group and within the worker group but incomplete between them, which is the only friction. The economy has a stochastic trend a_t , so all quantities are detrended by e^{a_t} before the model is solved; the detrended system is stationary in the states (x_t, σ_t^2, L_t) and $\xi_{i,t}$, and the Campbell-Shiller constants of Section 7.11 are evaluated at the detrended steady state, a fixed point of the log price-dividend ratio $\bar{z}$ that the solution computes.

A.12 Aggregation

Firms are equal-sized and of measure zero, so the aggregate is the equal-weight integral of the firm block. By (34) and (52), firm i 's value of hiring loads on the diagnostic aggregate and idiosyncratic beliefs $\tilde{x}_t$ and $\tilde{\xi}_{i,t}$,

$$pe_{i,t} = \bar{p}e + \Lambda(\tilde{x}_t + \tilde{\xi}_{i,t}), \quad (\text{A.7})$$

where $\tilde{\xi}_{i,t} = \xi_{i,t} + \zeta^\xi \sigma_\xi e_{i,t}$ is the firm's diagnostic idiosyncratic belief. Integrating over i and using $\int_0^1 \tilde{\xi}_{i,t} di = 0$ by the law of large numbers,

$$\int_0^1 pe_{i,t} di = \bar{p}e + \Lambda \tilde{x}_t, \quad (\text{A.8})$$

so the aggregate value of hiring loads only on the common diagnostic belief $\tilde{x}_t$, the constant discount $\bar{p}e$, and the idiosyncratic beliefs wash out. The cross-section is the dispersion of $\Lambda \tilde{\xi}_{i,t}$: the standard deviation across firms is $\Lambda \text{SD}(\tilde{\xi}_{i,t})$, which is the object matched to the cross-sectional dispersion of the value of hiring in Section 7.14. No firm-size distribution is needed: with equal-sized firms the aggregate is literally the mean firm. A size distribution would matter only for granular effects in which a few large firms move the aggregate, which this economy abstracts from.

A.13 Recursive internally rational equilibrium

Definition 1 (Recursive internally rational equilibrium). *Let the aggregate state be $\Omega_t = (a_t, x_t, \sigma_t^2, L_t)$ and the firm state add $\xi_{i,t}$. An equilibrium consists of a family value function and consumption policy $C^p(\Omega)$; firm values $J(\Omega, \xi)$ and a vacancy policy $v(\Omega, \xi)$; a wage function $W(\Omega, \xi)$; tightness $\theta(\Omega)$ and filling rate $q(\Omega)$; a stochastic discount factor $M(\Omega, \Omega')$; the subjective measure $\mathbb{F}$ of Section 7.7; and a perceived law of motion $\Gamma^\mathbb{F}$ and actual law of motion Γ , such that*

1. *given prices, dividends, and $\mathbb{F}$, the consumption policy solves the family problem (36) subject to the budget and goods-market clearing $C_t = e^{a_t} L_t + b e^{a_t} U_t - \kappa e^{a_t} v_t - F e^{a_t} L_t$, where the last term is the fixed operating cost of the active matches, and M satisfies (37);*
2. *given M , W , q , and $\mathbb{F}$, the vacancy policy solves the firm problem (40)-(41);*
3. *the wage W solves the Nash problem (47);*
4. *the matching function and (43) determine θ , q , and the law of motion of L ;*

5. *markets clear: labor by (43), goods by the clearing condition above, and equity by M pricing the family's claims;*
6. *the perceived law $\Gamma^{\mathbb{F}}$ is generated by $\mathbb{F}$ and the actual law Γ by the objective measure, and the two agree on all objects the family observes.*

A.14 State space and solution

State space and solution. The detrended economy has aggregate states (x_t, σ_t^2, L_t) and firm state $\xi_{i,t}$. The asset block is solved in closed form: the Epstein-Zin kernel and the loadings A_1, A_2, A_{1m}, A_{2m} of the consumption and dividend claims are the Campbell-Shiller fixed point of Appendix A.1, and the value of hiring is (52). The labor block is log-linearized: tightness is $\hat{\theta}_t = \alpha^{-1} \hat{P}_{i,t}$ from (42), employment follows the linearized (43), and the policy is solved by undetermined coefficients. The model is then simulated at the monthly frequency under the objective law of motion, with decisions taken under the subjective law $\mathbb{F}$; all reported moments are computed on the simulated path time-aggregated to the year, following the internal-rationality protocol of Adam and Marcet (2011).

A.15 The equity premium

The equity premium. The premium on any claim is $-\text{Cov}_t(m_{t+1}, r_{t+1})$. With the prices of risk (A.11) and the loadings of the value of hiring, the premium the family perceives is the constant

$$\mathbb{E}_t^{\mathbb{F}}[r_{t+1}^e - r_t^f] = (1 - \theta_{EZ})\kappa_1\kappa_{1m}A_1A_{1m}\varphi_e^2\bar{\sigma}^2 + (1 - \theta_{EZ})\kappa_1\kappa_{1m}A_2A_{2m}\sigma_w^2 \equiv \mathcal{K}\bar{\sigma}^2, \quad (\text{A.9})$$

constant because the family does not condition on σ_t^2 (Section 7.7). The prices of risk in $\mathcal{K}$ are the functions (A.11) of (γ, ψ) , not free. The premium *level* is set by scaling the pricing member's cash-flow exposure through s_D (Section 7.3), so it is calibrated to five percent and reported as such rather than as a prediction. Appendix A.1 reproduces the Bansal-Yaron (2004) benchmark from the same code as a validation gate for the solver, not for the calibrated economy. The mechanism that moves unemployment is not a time-varying premium but the belief: when the family's growth belief x_t falls, the long-duration value of hiring falls, vacancies fall, and unemployment rises.

A.16 The allocative versus the realized new-hire wage

The allocative wage and the realized new-hire wage. Pissarides (2009) objects that rigid-wage search models are refuted by the flexibility of realized new-hire wages (Haefke et al. (2013)). The model answers this because the object that matters for job creation is not the realized wage but the worker's *expected* outside option in the threat point (47), and it is that object the surveys measure to be acyclical. The realized new-hire wage is a different object: from (47) it is $\widehat{W}_t^{\text{new}} = \eta_w \widehat{A}_t + \eta_w \kappa_\theta \widehat{\theta}_t + (1 - \eta_w) \beta^w \widehat{A}_t$, where κ_θ is the steady-state elasticity of the search-cost term $\kappa e^{a_t} \theta_t$ in (47); the tightness term $\eta_w \kappa_\theta \widehat{\theta}_t$ is strongly procyclical, so the realized new-hire wage is far more flexible than the allocative wage $\omega = 0.097$. The model therefore reconciles a flexible realized new-hire wage, matching the evidence, with a rigid allocative wage, which is what generates the amplification, precisely the distinction Pissarides (2009) draws: the amplification requires rigidity in the *marginal* value of a job, not in the average or the realized wage, and the marginal value moves with the acyclical expected outside option.

A.17 The mechanism, step by step

The mechanism. It is useful to trace the two channels before the numbers. A positive growth surprise makes the family's diagnostic belief $\tilde{x}_t$ overshoot the fundamental x_t (49). Through the duration Λ (52) this raises the value of hiring $\widehat{P}_{i,t}$ by a large multiple, and because the dividend that finances the marginal investor's consumption rises with growth, the kernel prices this movement and delivers the equity premium and the volatility of valuations. The same rise in $\widehat{P}_{i,t}$ raises tightness through the job-creation condition (42); with a rigid wage the productivity-wage gap opens, operating leverage (Lemma 1) magnifies the profit, and vacancies rise and unemployment falls. Objectively the overshoot $\tilde{x}_t$ reverts to x_t , which reverts at ρ , so the elevated $\widehat{P}_{i,t}$ predicts low returns and the movement reads as discount-rate news (Proposition 2); subjectively the family, perceiving a constant return and

anchoring the long run to the truth, reads the same movement as near-term cash-flow news (Proposition 3). The growth belief carries the asset-price volatility and both decomposition shares; the unemployment volatility is carried by the rigid wage, through which the operating-leverage productivity-wage gap moves the value of a filled job.

A.18 The log kernel and prices of risk

Log-linearizing (37) around the balanced-growth path with the standard Campbell-Shiller approximation gives the log kernel innovation

$$m_{t+1} - \mathbb{E}_t^{\mathbb{F}}[m_{t+1}] = -\lambda_\eta \sigma_t \eta_{t+1} - \lambda_e \sigma_t \varepsilon_{t+1} - \lambda_w \sigma_w w_{t+1}, \quad (\text{A.10})$$

with prices of risk that are functions of preferences and the consumption loadings, not free parameters:

$$\lambda_e = \gamma \text{ (consumption shock } \varepsilon), \quad \lambda_\eta = (1 - \theta_{EZ}) \kappa_1 A_1 \varphi_e \text{ (long-run-risk shock } \eta), \quad \lambda_w = (1 - \theta_{EZ}) \kappa_1 A_2, \quad (\text{A.11})$$

where $A_1 = (1 - 1/\psi)/(1 - \kappa_1 \rho)$ and A_2 solve the wealth-consumption valuation (Appendix A.1) and κ_1 is the Campbell-Shiller linearization constant. The price of the short-run consumption shock is the coefficient of relative risk aversion; the price of the long-run-risk shock is the Epstein-Zin term that vanishes at $\gamma = 1/\psi$. Because C_t^p loads on the growth belief through s_D and the dividend, the same x_t that the family over-extrapolates enters the kernel and is priced.

A.19 The two measures

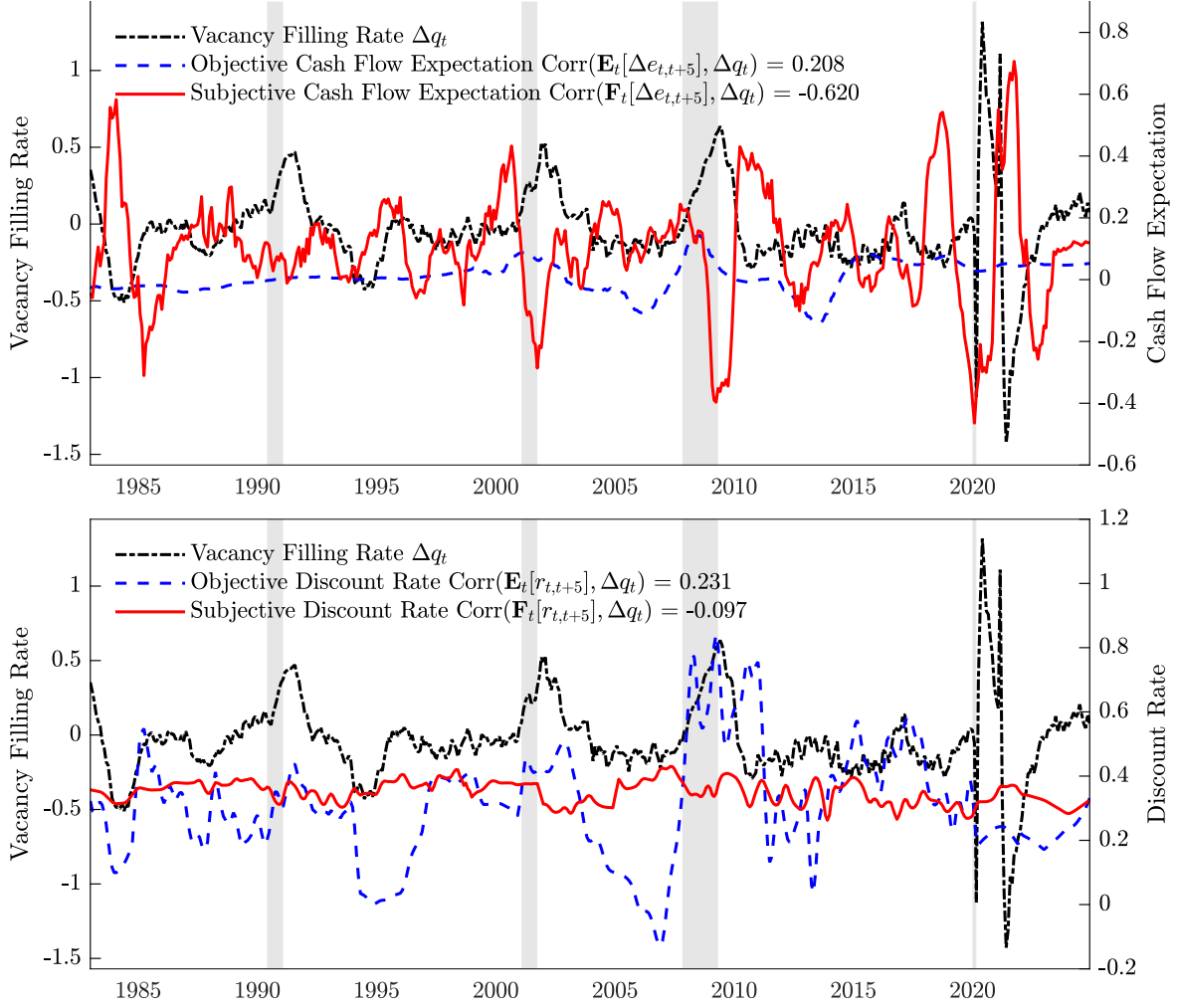
The two measures. The subjective measure $\mathbb{F}$ is generated by (49)-(50) and the perceived constant variance. The objective measure sets $\zeta = 0$ and restores the law of motion (33) of σ_t^2 . The two agree on every object the family observes, the history of growth, and differ only on the diagnostic distortion and the unobserved variance state, so $\mathbb{F}$ is a coherent probability measure and the equilibrium is internally rational in the sense of Adam and Marcet (2011). Because the diagnostic distortion is on the transitory innovation and not on the perceived persistence, the family running a persistence regression on its own long-run growth forecasts would find the correct ρ ; the overreaction it would detect is in the short-run revision, which is the object the Coibion-Gorodnichenko regression isolates.

B Appendix: Additional Results

B.1 Stylized Facts

Vacancy filling rate, discount rate, and expected cash flows Figure A.1 compares subjective and machine expectations for discount rates and cash flows, plotted against the vacancy filling rate. These series represent the components that drive hiring decisions in the search model. Machine expectations of discount rates exhibit a strong positive relationship with vacancy filling rates, particularly around recessions. This pattern aligns with the theoretical prediction that higher discount rates (reflecting greater compensation for risk) should coincide with lower hiring as firms perceive a lower present discounted value of employment. Survey expectations of discount rates, by contrast, are relatively flat and display little sensitivity to the business cycle, consistent with studies that find acyclical subjective risk premia (Nagel and Xu, 2022b). For cash flows, however, survey expectations show exaggerated cyclical variation, becoming sharply pessimistic during downturns when vacancy filling rates are high. Machine forecasts also vary cyclically but to a much lesser extent, consistent with survey respondents overreacting to new information when forming cash flow expectations.

Figure A.1: Vacancy Filling Rates, Discount Rates, and Expected Cash Flows

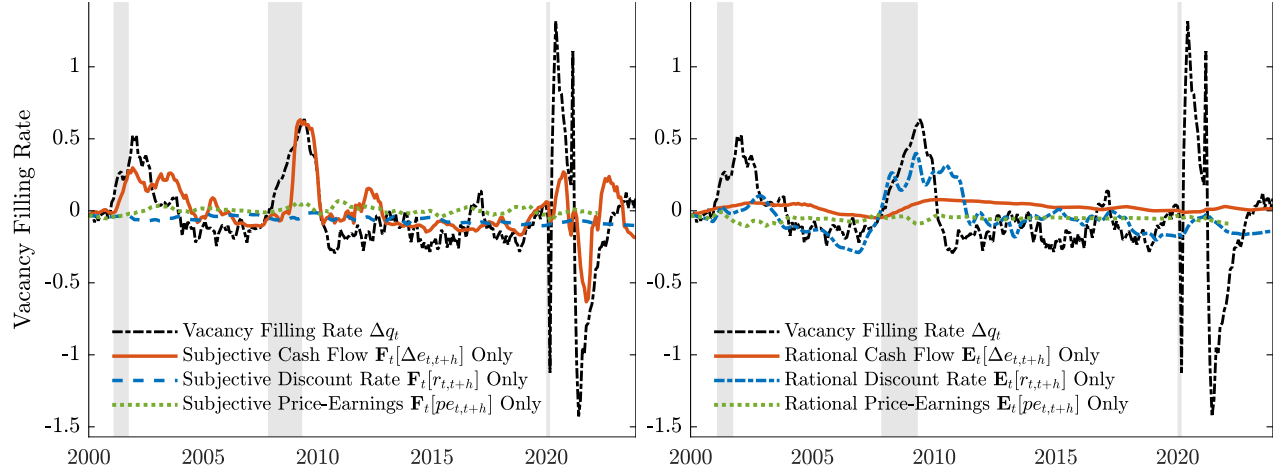


Notes: Figure plots $h = 5$ year ahead survey forecasts $\mathbb{F}_t[\cdot]$ and machine learning forecasts $\mathbb{E}_t[\cdot]$ of discount rates $r_{t,t+h}$ and annual cash flow growth $\Delta e_{t,t+h}$ (left axis) against the annual log growth in the vacancy filling rate q_t (right axis). x axis denotes the date on which each forecast has been made and the vacancy filling rate was realized. Subjective expectations $\mathbb{F}_t$ are based on survey forecasts from the CFO survey for stock returns, and IBES for earnings growth. Machine expectations are based on machine learning forecasts $\mathbb{E}_t$ from Long Short-Term Memory (LSTM) neural networks $G(\mathcal{X}_t, \beta_{h,t})$, whose parameters $\beta_{h,t}$ are estimated in real time using $\mathcal{X}_t$, a large scale dataset of macroeconomic, financial, and textual data. The out-of-sample forecast testing period is quarterly and spans 2005Q1 to 2023Q4. NBER recessions are shown with gray shaded bars.

B.2 Counterfactual Decomposition of the Vacancy Filling Rate

Figure A.2 plots counterfactual paths for vacancy filling rate growth driven by each component alone, using fitted values from the time-series decomposition and initialized to the actual 2005Q1 value. The subjective cash flow counterfactual tracks the realized path remarkably well, explaining much of its cyclical variation, whereas the objective discount rate component, though sizable, is visibly less pronounced.

Figure A.2: Role of Components in the Vacancy Filling Rate

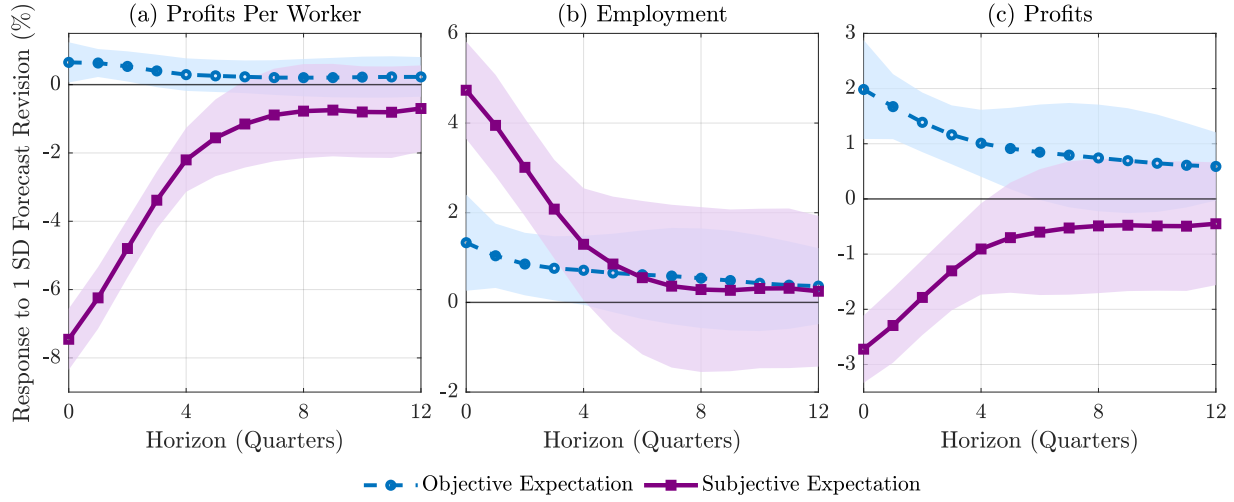


Notes: Figure presents counterfactual time series showing the evolution of vacancy filling rate growth if driven solely by each expectation component. Left panel shows subjective expectations; right panel shows objective expectations. Counterfactual series are constructed by accumulating fitted values from regressions of vacancy filling rate growth on individual expectation measures, with all series initialized to the actual vacancy filling rate growth in 2005Q1. Black line shows actual vacancy filling rate growth for comparison. Subjective expectations $\mathbb{F}_t$ are based on CFO survey forecasts (discount rates) and IBES analyst forecasts (cash flows, price-earnings). Objective expectations $\mathbb{E}_t$ are based on machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. Gray shaded areas indicate NBER recessions. Sample period: 2000Q4 to 2023Q4.

B.3 Employment Response to Forecast Revisions: Objective vs. Subjective Beliefs

Figure A.3 compares impulse responses under objective expectations (machine learning forecasts, blue dashed lines) versus subjective expectations (analyst forecasts, violet solid lines) across three related outcomes. Panel (a) shows that profits per worker remain stable under objective expectations following positive idiosyncratic shocks, while they decline significantly under subjective expectations. This contrast reveals that the decline in profits per worker documented in the main text is driven by belief distortions rather than rational responses to fundamental shocks.

Figure A.3: Employment Response to Idiosyncratic Forecast Revisions: Objective vs. Subjective Beliefs



Notes: Figure displays impulse responses to earnings growth forecast revisions, estimated via local projections: $y_{i,t+h} = \beta \text{ForecastRevision}_{i,t} + \alpha_i + \tau_t + \varepsilon_{i,t+h}$, where α_i and τ_t denote firm and time fixed effects. Panel (a): log profits per worker $\log(E_{i,t+h}/L_{i,t+h})$. Panel (b): log employment $\log(L_{i,t+h})$. Panel (c): log profits $\log(E_{i,t+h})$. Blue dashed line: impulse response under objective expectations, where forecast revision is the machine learning earnings growth revision $\mathbb{E}_t[\Delta e_{i,t+1}] - \mathbb{E}_{t-1}[\Delta e_{i,t+1}]$. Violet solid line: impulse response under subjective expectations, where forecast revision is the analyst earnings growth revision $\mathbb{F}_t[\Delta e_{i,t+1}] - \mathbb{F}_{t-1}[\Delta e_{i,t+1}]$. Observations weighted by each firm's market value. Shaded areas: two-way clustered 95% confidence intervals by firm and time. Data sample: 1999Q1–2023Q4.

B.4 Alternative Survey Measures of Subjective Cash Flow Expectations

The large role played by subjective cash flow expectations in explaining the vacancy filling rate holds more generally across alternative survey forecasts of earnings growth. Table A.1 re-estimates the subjective variance decomposition while replacing IBES survey forecasts of earnings growth with the corresponding forecast from the Bloomberg (BBG) and CFO surveys. The forecast horizon for the CFO survey has been limited to $h = 1$ year ahead and the sample covers a shorter period over 2002Q1 to 2019Q3 due to missing earnings growth forecasts in the CFO survey.

To summarize the alternative survey measures into a single series, I construct Filtered Investor (FI) expectations by extracting the common component of subjective cash flow beliefs using a Kalman filter. The latent state variable is defined as the h -month-ahead expected earnings growth, $S_t \equiv \mathbb{F}_t[\Delta e_{t+h}]$, which captures investors' subjective beliefs about future cash flows. The observation vector X_t contains survey measures of expected earnings growth over the next h periods from IBES, Bloomberg, and CFO surveys. The Kalman filter then estimates the latent S_t as the optimal linear combination of these noisy survey indicators $S_t = C(\Theta) + T(\Theta)S_{t-1} + R(\Theta)\varepsilon_t$, where C, T, R are matrices of the model's primitive parameters $\Theta = (\alpha, \rho, \sigma_\varepsilon)'$. ε_t is an innovation to the latent expectation that was unpredictable from the point of view of the forecaster. α is the intercept, ρ is the persistence, and σ_ε is the standard deviation of the latent innovation error. The Observation equation takes the form $X_t = D + ZS_t + Uv_t$, where h is a fixed forecast horizon. The observation vector X_t contains measures of survey expected cash flows from IBES, BBG, and CFO surveys over the next h periods. v_t is a vector of observation errors with standard deviations in the diagonal matrix U . Z and D are parameters that have been set to 1s and 0s, respectively. I use the Kalman filter to estimate the remaining parameters $\alpha, \rho, \sigma_\varepsilon, U$. Since some of our observable series are not available at all frequencies and/or over the full sample, the state-space estimation fills in missing values using the Kalman filter.

Table A.1: Variance Decomposition of Vacancy Filling Rate: Alternative Subjective Cash Flow Expectations

Horizon h (Years)	1	2	3	4	5
Subjective Expectations: $\log q_t = c_q + \mathbb{F}_t[r_{t,t+h}] - \mathbb{F}_t[e_{t,t+h}] - \mathbb{F}_t[pe_{t,t+h}]$					
(a) Filtered Investor (FI) Expectations					
(-) Cash Flow	0.578***	0.625***	0.684***	0.887***	0.933***
t -stat	(3.046)	(4.275)	(4.894)	(6.019)	(7.613)
N	76	76	76	76	76
(b) Bloomberg (BBG) Survey					
(-) Cash Flow	0.575***	0.686***	0.765***	1.054***	1.163***
t -stat	(7.274)	(8.200)	(7.705)	(5.815)	(5.112)
N	76	76	76	76	76
(c) CFO Survey					
(-) Cash Flow	0.637*				
t -stat	(1.934)				
N	71				

Notes: Table reports variance decompositions of the vacancy filling rate while replacing IBES earnings growth forecast with alternative surveys as measures of subjective cash flows. FI summarizes the alternative survey measures into a single series using a Kalman filter. The sample for BBG and FI is quarterly from 2005Q1 to 2023Q4. The sample for CFO is quarterly from 2002Q1 to 2019Q3. Newey-West corrected t -statistics with lags = 4 are reported in parentheses: *sig. at 10%. **sig. at 5%. ***sig. at 1%.

B.5 Financial Constraints

A natural concern is that variation in hiring may reflect differences in financial constraints rather than distortions in beliefs. In a rational expectations model, financial constraints appear as a Lagrange multiplier that tightens the firm’s stochastic discount factor (SDF), raising internal hurdle rates and suppressing hiring (Kehoe et al., 2019). In this setting, constraint-induced fluctuations in hiring would be objectively attributed to higher discount rates. By contrast, under subjective expectations, survey respondents may misattribute the effect of constraints to lower future cash flows, especially if internal hurdle rates are persistent, upward-biased, and unresponsive to market conditions (Gormsen and Huber, 2025). Financial constraints could also allow the effects of belief distortions to persist by limiting arbitrage that would otherwise correct them (De La O et al., 2024).

Measures of Financial Constraints To test these hypotheses, I incorporate firm-level financial constraint measures into the decomposition framework:

- **Firm Size (Total Assets):** Firms in the bottom tertile of the asset size distribution are classified as financially constrained, while those in the top tertile are unconstrained (Erickson and Whited, 2000).
- **Payout Ratio:** Defined as dividends plus stock repurchases scaled by total assets. Firms with the lowest (highest) payout ratios are classified as constrained (unconstrained), consistent with the idea that constrained firms conserve internal funds (Fazzari et al., 1988).
- **SA Index:** The size-age index developed by Hadlock and Pierce (2010), constructed as $SA = -0.737 \cdot \text{Size} + 0.043 \cdot \text{Size}^2 - 0.040 \cdot \text{Age}$, where Size is log real assets and Age is years since listing. Higher SA values indicate tighter constraints.
- **Expected Free Cash Flow:** Based on Lewellen and Lewellen (2016), firms are sorted into constraint groups using predicted free cash flow, estimated from cross-sectional regressions on lagged characteristics. Low expected FCF implies tighter constraints.
- **WW Index:** The Whited-Wu index (Whited and Wu, 2006), a linear combination of cash flow, dividend status, leverage, size, and sales growth, where higher index values imply greater constraints.

Each measure is updated annually and firms are classified based on tertiles or continuous index values. Each measure is aggregated to the portfolio level and standardized before entering the regression as controls.

Decomposition with Financial Constraints I modify the baseline decomposition regression as follows:

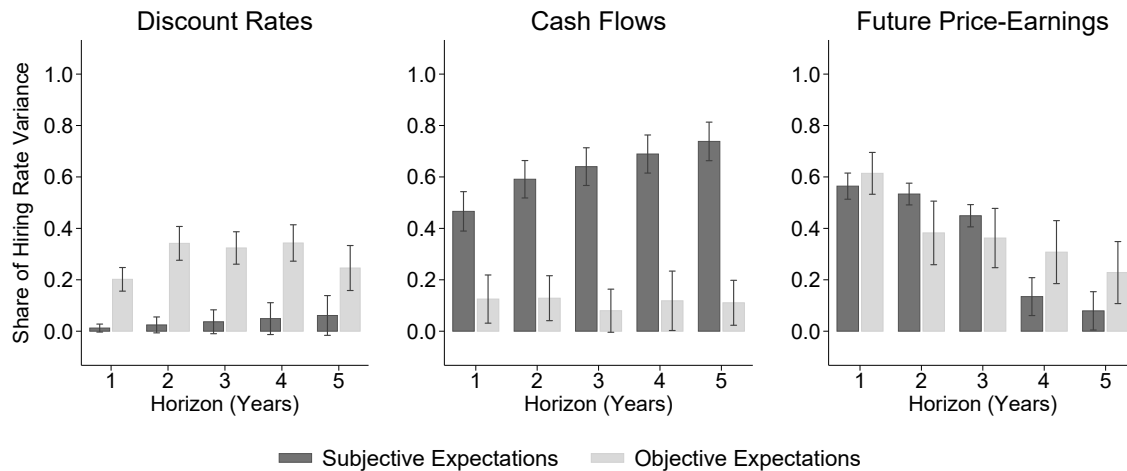
$$\mathbb{F}_t[y_{i,t,t+h}] = \beta \cdot hl_{i,t} + \Gamma \cdot \text{FC}_{i,t} + \alpha_i + \alpha_t + \varepsilon_{i,t}$$

where $y = r, e, pe$ denotes either discount rates, cash flows, or price-earnings (the components of the decompositions), and $\text{FC}_{i,t}$ is a vector of standardized financial constraint measures for firm i at time t . α_i and α_t are firm and time fixed effects. As before, the parameter of interest is β , which captures the share of variation in the hiring rate $hl_{i,t}$ explained by subjective expectations, but this time conditional on financial constraints. I run analogous regressions to estimate the contributions of discount rate expectations and future price-earnings ratios. I also replace survey forecasts with machine learning forecasts to estimate the decomposition under objective expectations, again controlling for financial constraints using the same specification.

Results Figure A.4 presents the decomposition estimates with and without financial constraint controls, under both subjective and objective expectations. Under subjective expectations, the contribution of expected earnings to hiring variation remains large and significant, with only a modest reduction in explanatory power after controlling for financial constraints. This suggests that distorted beliefs about cash flows persist even after adjusting for observable constraint-related fundamentals. These findings are consistent with the view that constrained firms overreact to cash flow news or internalize persistent pessimism about earnings. Under objective expectations,

however, the contribution of discount rate expectations drops substantially once constraint controls are included. This is consistent with a rational model in which financial constraints tighten the SDF and raise internal hurdle rates. When this variation is accounted for, the rational model assigns less importance to discount rate news in explaining hiring variation. The result supports the interpretation that financial constraints can explain a non-trivial share, but do not fully explain, variation in hiring. While objective forecasts attribute constraint effects to discount rates, subjective expectations appear to reflect persistent pessimism about cash flows.

Figure A.4: Cross-Sectional Decomposition of Hiring Rate: Control for Financial Constraints



Notes: Figure estimates cross-sectional decomposition of hiring rate, controlling for measures of financial constraints. Light bars show contributions under objective expectations; dark bars show contributions under subjective expectations. Financial constraint controls include firm size, payout ratio, SA index, expected free cash flow, and the Whited-Wu index. The sample is quarterly from 2005Q1 to 2023Q4. Each bar shows Newey-West 95% confidence intervals with lags = 4.

Data Availability Statement

The data and code that support the findings of this paper are provided in a replication package. The analysis draws on publicly available sources (FRED and BLS macroeconomic series, the Job Openings and Labor Turnover Survey, the Federal Reserve Bank of Richmond and Duke University CFO Survey, the Survey of Professional Forecasters, and NBER recession dates) together with restricted-access data obtained through Wharton Research Data Services (IBES analyst forecasts, CRSP prices and returns, and Compustat fundamentals), Value Line target prices, and licensed *Wall Street Journal* text used to construct the machine-learning forecasts. The proprietary data cannot be redistributed under their licenses; the replication package documents each source and provides all public data together with the code needed to reproduce the results for researchers with the appropriate subscriptions.